Is the Putative Mirror Neuron System Associated with Empathy? A Systematic Review and Meta-Analysis.

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Abstract

Theoretical perspectives suggest that the mirror neuron system (MNS) is an important neurobiological contributor to empathy, but empirical support is mixed. Here, we adopt a summary model for empathy, consisting of motor, emotional, and cognitive components of empathy. This review provides an overview of existing empirical studies investigating the relationship between putative MNS activity and empathy in healthy populations. 52 studies were identified that investigated the association between the MNS and at least one domain of empathy, representing data from 1044 participants. Our results suggests that emotional and cognitive empathy are moderately correlated with MNS activity, while motor empathy showed no relationship. Results varied across techniques used to acquire MNS activity (TMS, EEG, and fMRI). Overall, results provide some evidence for a relationship between the MNS and empathy. Our findings also highlight methodological variability in study design as an important factor in understanding this relationship. We discuss limitations regarding these methodological variations and important implications for clinical and community translations, as well as suggestions for future research.

Keywords: Mirror neurons; mirror neuron system; inferior frontal gyrus; inferior parietal lobule; action-observation; biological motion; empathy; simulation; mimicry; imitation; facial recognition; emotional resonance; emotional contagion; theory of mind (ToM); perspective-taking; mentalising; predictive coding; associative learning; TMS; motor resonance; EEG; mu suppression; fMRI; meta-analysis; systematic review.

1. Introduction

Empathy is a broad concept that refers to the reactions of an individual in response to the experiences of an observed other (Davis, 1994). Empathy is thought to be essential for effective social functioning, for instance in developing social understanding, maintaining interpersonal relationships, and facilitating pro-social behaviour. Empathic abilities putatively confer evolutionary advantages, such as the development and maintenance of healthy social relationships, reproductive advantage due to enhanced intimate partnerships, increased parental care and child-rearing ability, and the formation of organised social groups and communities (Decety, Norman, Berntson, & Cacioppo, 2012). Notably, the positive outcomes associated with empathy are thought to occur as a result of multiple important processes. It is the amalgamation of these processes that is thought to lead to empathic responses, which can facilitate positive prosocial outcomes such as compassion, altruism, and healthy interpersonal relating (Baron-Cohen, Lombardo, & Tager-Flusberg, 2013). As such, phenomenological accounts of empathy must account for numerous processes including the capacity to understand the thoughts and feelings of others, the ability to experience them, and the capability to respond in a caring and prosocial manner (Dvash & Shamay-Tsoory, 2014).

Research in the field of cognitive neuroscience has identified the mirror neuron system (MNS) as a potentially crucial neural substrate for empathy. However, a number of methodological and conceptual issues have emerged that raise questions regarding the nature of the MNS-empathy relationship. For example, differences in definitions and measurements of empathy adopted in prior studies have contributed to inconsistent outcomes relating to the role of the MNS in empathy. Further, while there are numerous ways to measure MNS activity, a ‘gold standard’, outside of direct neural recording, has not yet been established, and methods of eliciting a mirror neuron response vary across studies, also resulting in inconsistencies.

Despite intuitive theoretical propositions as to the important role that the MNS might play in empathy, there is not yet any definitive survey of the literature to provide empirical evidence for these claims. Here, we first review the construct of empathy and its underlying processes. Next, we discuss empathy as a multi-component construct. We then provide an overview of the MNS and its neural measurement, followed by the theoretical accounts for its potential role in empathy. Finally, we present the purpose and aims of the current systematic review and meta-analysis to address limitations in the literature.

1.1. Conceptualisations of Empathy and its Measurement

Empathy has traditionally been defined as a singular process, however recent evidence suggests that empathy is a more complex construct that is multidimensional in nature, consisting of a number of underlying processes that collectively facilitate the experience of empathy (Davis, 1994; Preston & de Waal, 2003). As such, empathy can be considered an umbrella term consisting of independent yet interactive components that, at a broad level, comprise of *motor* (mimicry or imitation), *emotional* (emotional resonance, and/or emotional contagion), and *cognitive* (i.e., theory of mind [ToM], mentalising, and/or perspective taking) components. Below, we provide a narrative overview of these subcomponents and the methodological approaches typically used in the literature.

1.1.2. Motor Empathy

Motor empathy can be defined as the automatic mimicry and synchronization of expressive body language, such as facial expressions, during the observation of another (Dimberg, 1990). This process occurs predominantly on a subconscious or automatic level, and is considered a lower-order, primitive behaviour that is evident early in development (Brook & Kosson, 2013). Throughout early development, bidirectional relationships between mental/emotional states and motor behaviours are thought to be forged. In turn, these are thought to inform the ability to make inferences about the internal experiences of others via a

putatively developed ‘map’ of associations between physical action and corresponding mental and emotional experiences (Meltzoff & Decety, 2003).

Motor empathy can be measured in multiple ways, for example via assessment of a person’s facial muscle activity during the observation of facial expressions in others. This is typically assessed using action coding systems (such as the facial action coding system (FACS; Ekman & Friesen, 1977), the mature imitation task (Rogers, Cook, & Greiss-Hess, 2005), the facial expression coding system (FACES; Kring & Sloan, 2007), or via facial electromyography (fEMG). When using action coding systems, trained independent coders identify and document observable changes in gestures and/or facial expressions in observed others. That is, facial muscle movements or contractions of participants during the presentation of a stimulus are observed by coders and recorded as consistent with a facial expression (e.g., a smile) based on a coding manual. One common measure using action coding system is the FACS. This measures motor empathy using a comprehensive anatomical coding system that assesses contractions of 44 different muscles during the observation of facial expressions, and has been shown to have good inter-rater reliability. For example, viewing happy or positively valanced images is associated with increased electrical activity of the zygomaticus major (cheek) muscle region (i.e., muscle movements involved in smiling). Conversely, contractions in the corrugator supercillii (eyebrow) muscle region that wrinkles the eyebrows into an angry or negative expression, is correlated with exposure to sad and/or angry faces, as well as unpleasant images (Dimberg, Andréasson, & Thunberg, 2011). Similarly, the FACES task presents a number of facial expressions and coders provide information pertaining to the valence (positive or negative), frequency, intensity, and duration of changes in facial movements of the observer (Kring & Sloan, 1991). This is acquired using a coding system, where expressions are defined as movement changes relative to ‘neutral’ facial displays (Kring & Sloan, 1991). The mature imitation task requires imitation of facial expressions and gestures following observation of actors, and the accuracy

of imitative abilities are recorded and assessed (Rogers et al., 2005). Comparably, fEMG assess changes in facial expressions via surface electrodes, and is a more sensitive measure. Facial EMG is able to detect very brief and subtle contractions of facial muscle fibres (reflecting facial muscle activity) that occur below the visual detection threshold (Mauss & Robinson, 2009).

It is suggested that mimicking and imitating motor behaviours (i.e., motor empathy) provides a gateway through which we can understand the thoughts and feelings of others (Iacoboni, 2009). However, while motor empathy is considered necessary to provide a strong foundation from which empathy later develops, it alone is not sufficient to achieve the full capacity to empathise (Decety & Meyer, 2008). To achieve the full capacity to empathise, a shared emotional experience between the perceiver and observed agent, as well as a cognitive understanding of another, is thought to be important.

1.1.3 Emotional Empathy

Emotional empathy can be defined as the ability to immediately detect and resonate with the emotional state of another person (Decety & Jackson, 2004). From a process perspective, it is thought that detecting and accurately recognising the emotional state of another person can lead to physiological changes within the observer that, if accurate, are congruent with the emotional state of the observed other. This ability to experience a synchronous psychophysiological state following observation is thought to help individuals converge emotionally with others, and assist in the ability to understand another (Shamay-Tsoory, 2011). The degree of emotional matching or sharing between the perceiver and the observed agent is thought to be dependent upon the existing internal representations within the perceiver. According to Preston (2007), it is improbable that the emotional states of the perceiver and observed other will completely match, rather the resonance between the two lies on a continuum and is dependent upon previous experience, similarity, and familiarity with the observed other (Preston & de Waal, 2003). In addition, while activation of internal

representations within the perceiver is automatic, the perceiver must attend to the emotional state of the observed other (Preston & de Waal, 2003). Further, in order to successfully empathise, the observer must not only be attentive, but also must also maintain self-other distinctions, and inhibit contagious unwanted emotions, such as distress, when perceiving and sharing emotions of the observed agent (Preston, 2007; Preston et al., 2007).

Emotional empathy can be measured objectively by providing insights into physiological changes experienced within the observer. Physiological changes in the autonomic nervous system following observation of another's emotional state is typically considered a reflection of emotional experience. Basch (1983) theorised that a perceiver's tendency to generate an identical psychophysiological emotional state to an observed target occurs as a result of similar biological programming of their respective autonomic nervous systems. That is, this 'physiological linkage' between the perceiver and observed agent occurs due to commonality of emotional experience, and reliable "mapping" of physiological patterns of activity onto emotional states or affects (Basch, 1983; Levenson & Ruef, 1992). These physiological changes can be measured in a number of ways, including assessment of electrodermal activity (EDA) and cardiovascular activity (ECG). Electrodermal activity (also referred to as galvanic skin response [GSR]) refers to variations in the electrical activity of the skin. This occurs when the sympathetic nervous system becomes increasingly active in response to emotional stimuli, resulting in a sweat gland activity upsurge reflecting emotional arousal (Kreibig, 2010; Neumann & Westbury, 2011). Similarly, cardiovascular reactivity (or heart rate [HR]) has been considered indicative of autonomic activation during emotional stimuli. Heart rate has been suggested to be reflective of emotional responses to other people's distress and a sensitive measure of emotional states (Kassam & Mendes, 2013; Kreibig, 2010; Neumann & Westbury, 2011). Studies obtaining physiological measures of empathy have typically also included self-report measures to provide a rich and comprehensive measurement of emotional empathy. The use of both objective and subjective

measures allows for a robust and clear measurement of emotional responsiveness, as opposed to simply measuring physiological arousal. Furthermore, subjective measures of empathy provide phenomenological data, accessing participants' personal insight and perceptions regarding their ability to empathise. The most commonly used questionnaires include the interpersonal reactivity index (IRI; Davis, 1980) and the empathy quotient (EQ; Baron-Cohen & Wheelwright, 2004). However, self-report measures of emotional empathy often overlap with other empathy domains (particularly cognitive empathy). Nonetheless, these self-report questionnaires provide a specific emotional empathy subscale, which assesses the emotional component of empathy via statements such as "*I tend to get emotionally involved with a friend's problems*" (EQ; Baron-Cohen & Wheelwright, 2004) and participants are required to respond on a Likert scale. In addition to these, there are also self-report measures available that assess various pathological traits of certain disorders that tap into the subcomponents of empathy, such as the Toronto alexithymia scale-20 (TAS-20; Taylor, Ryan, & Bagby, 1985), where participants are required to respond to questions such as "*I prefer to talk to people about their daily activities, rather than their feelings"* on a 5-point scale.

In summary, emotional empathy can be conceptualised in its most basic form as the ability to immediately detect and resonate with the emotional state of another person (Decety & Jackson, 2004). This contrasts with motor empathy, which is focused on simple mimicry of the motor expressions of another person. However, successful empathic responses require more than solely simulating and experiencing similar emotional states to an observed other (i.e., "*I feel what you feel*"), it also requires a *cognitive* perspective taking ability (i.e., "I understand what you feel"; Shamay-Tsoory, Aharon-Peretz, & Perry, 2009). Nevertheless, obtaining a shared emotional experience with an observed target is theorised to facilitate the ability to cognitively understand and infer the mental states of others, such as intentions, emotions, beliefs, and motivations (Decety & Meyer, 2008; Meltzoff & Decety, 2003).

1.1.4. Cognitive Empathy

Cognitive empathy (also known as ‘mentalising’ and ‘theory of mind’) is the final, broad component of empathy that putatively involves higher-order top-down mental processing. Cognitive empathy can be defined as the ability to comprehend the thoughts, feelings, beliefs, and intentions of others (Frith & Frith, 1999). This includes understanding the mental and emotional world of another person, and has been described as the ability to put one’s self into another’s shoes (i.e., ‘perspective taking’; Baron-Cohen et al., 2013). Cognitive empathy is thought to be facilitated by information that is acquired via motor and emotional empathic processes, which putatively helps the observer gain a strong understanding of the mental world of another and assisting in the ability to predict another’s intended behaviour.

Cognitive empathy is typically assessed by examining participants’ ability to identify and comprehend the mental and emotional states of an observed other (Baron-Cohen et al., 2013; Frith & Frith, 1999). This is achieved through objective tasks such as variations of the facial emotion recognition task (FERT; Montagne, Kessels, De Haan, & Perrett, 2007), the reading the mind in the eyes task (RMET; Baron-Cohen, Wheelwright, Hill, Raste, & Plumb, 2001) and the awareness of social inference test (TASIT; McDonald et al., 2004). Performance (typically assessed in terms of accuracy and/or reaction time) in these tasks provide an index of the ability to recognise emotions, identify mental states, and measure sensitivity to conversational inference using contextual information, respectively (Baron-Cohen et al., 2001; McDonald et al., 2004; Montagne et al., 2007), and collectively provides insight into cognitive empathy capacities. Similar to emotional empathy, cognitive empathy can also be assessed subjectively and the most commonly used self-report measures include the IRI (Davis, 1980) and EQ (Baron-Cohen & Wheelwright, 2004). Participants are provided with statements of various situations such as “*I sometimes find it difficult to see things from the ‘other guy’s’ point of view*” (IRI; Davis, 1980) and respond according to a Likert scale.

In summary, self-report measures of cognitive and emotional empathy are typically combined and assessed in unison, perhaps because they are considered elements of a single broader construct (i.e., empathy). These frequently used self-report questionnaires consist of both cognitive and emotional subscales, and some studies may opt to specifically measure only one subscale (i.e., only one specific component of empathy), while others prefer to assess total scores as a reflection of overall empathic abilities.

1.2. A Multi-Component Model for Empathy

Notably, the motor, emotional, and cognitive domains of empathy have been suggested to cohere into a unified, multi-process model that collectively produces the experience of empathy (Blair, 2005; Preston & de Waal, 2003). Despite this, studies have tended to adopt only one or two of these domains when defining and measuring 'empathy'. It has been previously stated that "There are probably nearly as many definitions of empathy as people working on the topic" (De Vignemont & Singer, 2006, p. 435; Gallese, 2008). This lack of consensus leads to methodological inconsistencies when measuring the same construct (i.e., 'empathy'), which resulting in mixed and often conflicting findings. Efforts to understand empathy in clinical populations and to develop new interventions, as well as attempts to train or augment empathic abilities in the community (e.g., health professionals for improved patient outcomes (Bearman, Palermo, Allen, & Williams, 2015), are likely to be impacted by disagreement within the literature. As such, a comprehensive, multi-process, and valid conceptualisation of empathy is imperative for future research.

Here, our approach to empathy is multidimensional, which adopts each of the components discussed earlier. In our multidimensional summary model, empathy can be summarised as a construct consisting of three key components, namely, motor, emotional, and cognitive processes, which functionally interact to achieve optimal capacity to effectively empathise. This model is presented diagrammatically in *Figure 1*. It should be noted that this summary model reflects early processes of empathy, and does not take into consideration

'outcomes' of empathy that may follow these processes, such as compassion. Each of the subcomponents of our model consist of specific interactive processes that are thought to be driven by a similar mechanism. While these processes have been considered relatively independent, they may interact to facilitate the ability to empathise, and are theorised to each be driven by a common enabling mechanism. One theorised functional mechanism has been referred to as *simulation* (Gallese & Goldman, 1998), whereby an implicit association is established between the internal state of an observer and that of the observed agent (Gallese, 2003). Simulation is thought to occur subconsciously and provide the ability to recruit an individual's own internal state and experience the internal states of others, enabling an individual to experientially share meanings of motor behaviours, emotions, and mental worlds (Gallese, 2008).

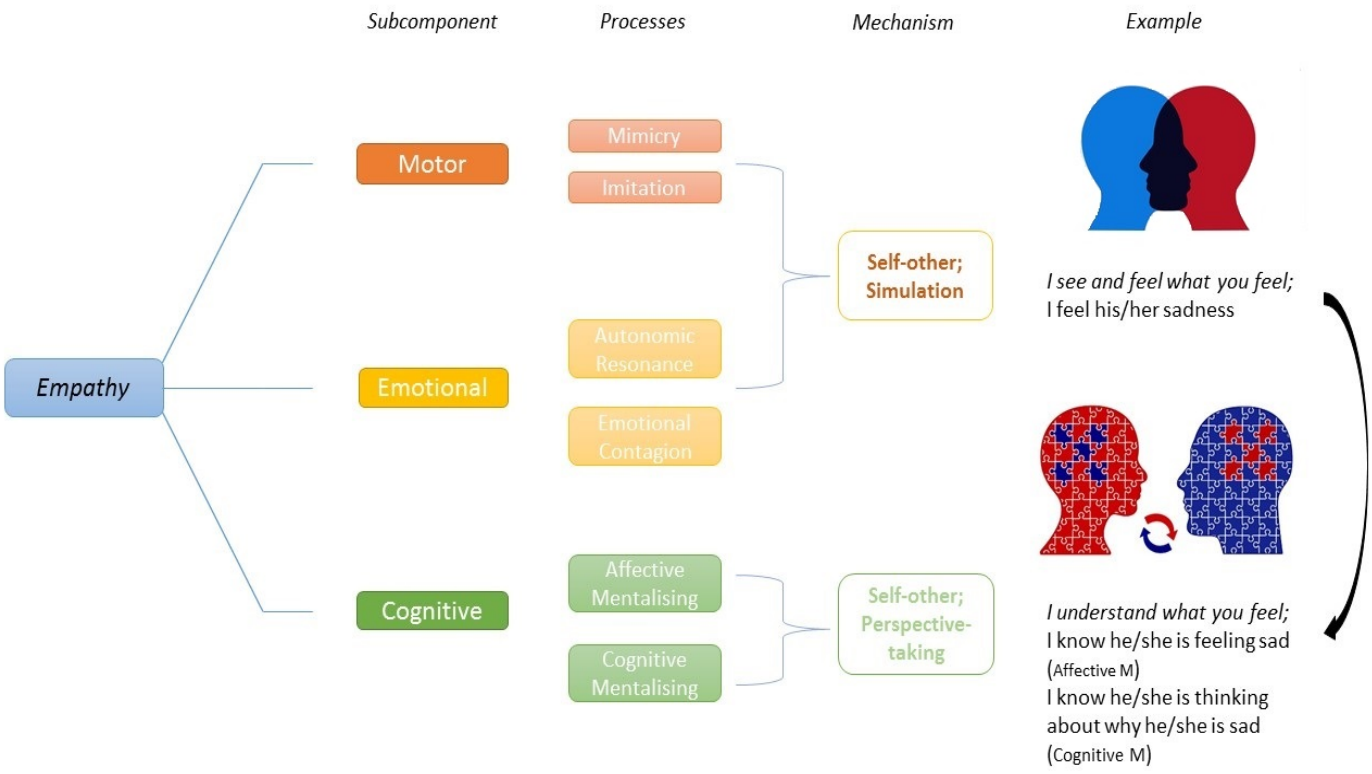


Figure 1- *An illustration of the multidimensional model for empathy summarised and adopted here. This model consists of motor, emotional and cognitive components that each consist of their respective processes. Motor empathy is a process of mimicry or imitation of observed motor behaviour, while emotional empathy is a process of autonomic and emotional resonance experienced within the self during observation of another's emotional display. Cognitive empathy is a process of identifying and understanding the cognitive and affective mental state/s of another. These processes are thought to be enabled via the same a simulation mechanism, whereby internal experiences are made reference to, in order to stimulate the experience of empathy.*

Aside from matters pertaining to the definition and measurement of empathy, there are current attempts to understand the neurological basis of this construct. A number of brain networks, collectively termed the 'social brain', have been implicated in empathy and social cognition as a result (Kennedy & Adolphs, 2012; Bernhardt and Singer (2012); (Kennedy & Adolphs, 2012). One of the most controversial debates regarding the neural underpinnings of empathy concerns the potential role of the mirror neuron system (MNS).

1.3. The Mirror Neuron System

Mirror neurons were first discovered in area F5 of the premotor region in macaques via single-cell recordings (Di Pellegrino, Fadiga, Fogassi, Gallese, & Rizzolatti, 1992). Of these neurons, a subclass discharged during both the execution of action *and* during the observation of action, reflecting what is now termed 'mirroring' (Rizzolatti & Craighero, 2004). Mirror neurons have since been demonstrated to be specialised cortical cells that fire during both the execution of an action and during the observation of a similar action (Rizzolatti & Craighero, 2004). Areas homologous to these 'mirroring' regions discovered in the macaque monkey have now been established in humans through neuroimaging and electro-physiological research (Rizzolatti, Fadiga, Matelli, et al., 1996). These regions, now collectively referred to as the mirror neuron system (MNS; also commonly referred to as the 'core' MNS; visuo-motor mirror neuron system [VM-MNS], the parieto-frontal MNS [PF MNS], fronto-parietal MNS [FP MNS], and the action observation network [AON]; Rizzolatti & Craighero, 2004; Rizzolatti, Fadiga, Gallese, & Fogassi, 1996), include the inferior frontal gyrus (IFG) and the inferior parietal lobule (IPL; Rizzolatti & Craighero, 2004). In terms of cortical hierarchy within the MNS, it is understood that visual sensory information enters through the visual cortex and is coded by the superior temporal sulcus (STS). This information is then transferred via anatomical connections to the IPL, where kinaesthetic aspects of observed actions are coded and then forwarded to the IFG, where

goals of actions are coded for and circulated back to the STS (Carr, Iacoboni, Dubeau, Mazziotta, & Lenzi, 2003). It should be noted that the STS is considered an associated region of the MNS that is important for sensory input (i.e., fires only during observation of movement), codes for visual descriptions of actions, and is critical for processing biological motion, but does not actually contain mirror neurons (Carr et al., 2003), unlike the IFG and IPL (Kilner & Lemon, 2013). Further to this, an "extended MNS" has also been proposed based on the understanding that the 'core' MNS (or VM-MNS) communicates with other networks important for emotion and higher-order processing (e.g., the limbic system; Carr et al., 2003; Cattaneo & Rizzolatti, 2009; Pineda, 2008). While this extended MNS does not necessarily contain mirror neurons, it is considered an anatomical extension which may functionally interact with VM-MNS (Carr, Iacoboni, Dubeaut, Mazziotta, & Lenzi, 2003; Pineda, 2008).

1.3.1. Measuring the Mirror Neuron System

A number of methods have been developed to measure mirror neuron activity in humans. Because mirror neurons cannot be directly measured in humans non-invasively (i.e., via single-cell recordings, as assessed in macaque monkeys), mirror neuron activity can generally only be indirectly measured. Additionally, these techniques require that mirror neuron activity is elicited through action observation; that is, presenting stimuli of human motor actions such as finger movements, hand-grasping actions, facial expressions, or biological motion (Iacoboni, 2009; Rizzolatti & Craighero, 2004). Currently, there are three commonly used neuroscientific methods for measuring mirror neuron activity. The first method involves using functional magnetic resonance imaging (fMRI), where changes in blood-oxygenation level dependent (BOLD) responses in areas of the MNS during action observation, relative to a control condition, are considered reflective of mirror neuron activity (Rizzolatti & Craighero, 2004). Another method assesses mu-suppression using electroencephalogram (EEG). This method is a measure of resting motor neurons, where

large-amplitude EEG oscillations in the mu frequency band (8-13 Hz) over scalp regions CZ, C3, and C4 (or, F3, F4, and FCz; Muthukumaraswamy, Johnson, & McNair, 2004; Oberman et al., 2005) are typically observed during rest due to the spontaneous and synchronized firing of sensorimotor neurons. However, when motor neurons are activated during the observation or execution of an action, this firing becomes suppressed/desynchronized. This suppression (mu suppression) of the sensorimotor cortex during the observation of movement is thought to reflect fluctuations in mirror neuron activity (Muthukumaraswamy et al., 2004; Oberman et al., 2005). Recent primate research also indicates that, during action observation, neuronal activation in the ventral premotor cortex (vPMC) is associated with mu suppression recorded over central electrodes, suggesting that motor and mirror activity contribute to mu-rhythm in the beta-band (Bimbi et al., 2018). Finally, mirror neurons can be assessed through changes in corticospinal excitability (CSE) via administration of single pulse transcranial magnetic stimulation (sTMS) to the primary motor cortex (M1). TMS is a non-invasive technique of neuromodulation, where electrical currents are induced in the brain by passing magnetic pulses through the skull (Pascual-Leone, 1999). sTMS to M1 produces a muscle response (referred to as a "motor evoked potential" [MEP]), which is considered an index of CSE (Fadiga, Fogassi, Pavesi, & Rizzolatti, 1995; Strafella & Paus, 2000). When this method is implemented during observation of visual stimuli illustrating execution of the same muscle targeted, an increase in cortical excitability (i.e., increased MEP or muscle response) typically occurs. This increase in MEP amplitude during action observation, relative to a control condition, is considered to reflect mirror neuron activity occurring in the adjacent and anatomically linked premotor cortex via inputs to the primary motor cortex, and is also referred to as "interpersonal motor resonance" (Fadiga et al., 1995; Gangitano, Mottaghy, & Pascual-Leone, 2001; Strafella & Paus, 2000).

Given the number of techniques available to measure MNS activity, methodological issues are important to consider when understanding their validity. In particular, the degree to

which mirror neuron responses can be affected by the type of stimulus that is presented is unclear. Specifically, studies have used both natural (i.e., human) and stylized (i.e., animated/human-like) stimuli. Regarding type of stimuli, it has been previously theorized that the use of unnatural (or, 'stylized') images potentially elicits different neurological responses and behavioural outcomes relative to natural images (Mori, 1970; Sarkheil, Goebel, Schneider, & Mathiak, 2012). This theory was originally presented as the 'uncanny valley' phenomenon, where the unfamiliarity of human-like presentations, such as avatars or robots, evokes within the observer feelings of 'eeriness', and has been reported in a number of studies involving androids and virtual characters (Mori, 1970). In particular, non-human (or 'dissimilar human') stimuli are thought to be processed in different ways, both behaviourally and neuronally, relative to the presentation of natural human faces or actions (Sarkheil et al., 2012). It is understood that the feeling of 'eeriness' is elicited when the stimulus is almost, but not quite, human. That is, when the presented humanoid stimulus (such as an avatar) looks increasingly human-like, whilst also containing non-human features (Mori, 1970). However, the current literature is somewhat mixed in this regard, with some studies showing differential brain activity in areas including the premotor cortex and fusiform gyrus, relative to natural human stimuli (de Borst & de Gelder, 2015; Tai, Scherfler, Brooks, Sawamoto, & Castiello, 2004), while others have found that the use of avatars evoke similar brain activation in certain regions such as the amygdala (Moser et al., 2007). It has also been demonstrated that perception of dissimilar human faces lead to behavioural differences (such as in emotion recognition, perceptual discrimination and emotional intensity ratings). However, these differences seem to be contingent upon the level of dissimilarity of the observed agent. That is, as the stimuli presented become increasingly human-like (e.g., from robots to human-like avatars), these behavioural differences begin to diminish (de Borst & de Gelder, 2015). Though, some differences in brain activity do remain, and this is thought to be due to differences in physical properties between human and non-human stimuli (de Borst &

de Gelder, 2015). The current understanding of whether there are any effects on eliciting MNS activity is unclear and has not yet been empirically tested or accounted for.

Another stimulus feature that may affect the magnitude of mirror neuron activity elicited is whether the image is dynamic (e.g., moving images or videos) or static (e.g., still images). It is generally thought that the use of dynamic stimuli may be more robust for generating MNS activity, as it is considered more ecologically valid and involves biological motion. Some studies report heightened activity in a number of brain structures involved in social cognitive processes, including the MNS, during the presentation of dynamic (relative to static) images (Rymarczyk, Żurawski, Jankowiak-Siuda, & Szatkowska, 2016; Trautmann, Fehr, & Herrmann, 2009). A more recent examination further supported this, where dynamic stimuli were found to be superior in evoking facial mimicry and neural network activations relative to static images (Rymarczyk, Żurawski, Jankowiak-Siuda, & Szatkowska, 2018). However, some studies have also shown elicitation of MNS activity using static images (Flournoy et al., 2016; Hooker et al., 2008; Schulte-Rüther et al., 2007). The question of whether these variations in methods have differential effects remains unclear.

TMS and EEG may also produce divergent MNS-related results, even when attempting to measure the same system (for example, Lepage, Saint-Amour, & Théoret, 2008). This particular study found that, although both measures sufficiently and reliably induced a mirror neuron response, there was no association found between MEP amplitude acquired through TMS and mu rhythm modulation during EEG (Lepage et al., 2008). These methodological uncertainties suggest that the field is in need of further progress and investigation towards establishing more reliable and effective methods of both eliciting and measuring MNS activity, and should aim to develop clear, robust, and empirically supported procedures to be broadly implemented. In addition, studies examining the MNS need to account for these methodological differences, as this may have important implications.

1.4. The Role of the Mirror Neuron System in Empathy

The role of the MNS in empathy remains inconclusive, and largely based on theoretical models. One key question relates to how movement observation, which leads to MNS activation, can in turn lead to experiencing empathy and understanding of the intentions of others. Suggestions that the MNS may provide important neural mechanisms for empathy were discussed early in the literature (Gallese, 2001; Gallese & Goldman, 1998). Gallese (2001, 2003) posited that mirror neurons provided an '*as if*' mechanism, where the motor system becomes active during the observation of another's action, without actually producing the observed action itself. In other words, perceiving an action was thought to correspond to internally simulating it, which he termed 'embodied simulation', providing an implicit link between the observer and the observed agent. This implicit ability provided by mirror neurons was described as an unconscious and automatic process, and argued to be a suitable mechanism for enabling empathy and social bonds (Gallese, 2003). Further to this, the 'shared-manifold' hypothesis was proposed, postulating that shared neural systems and representations are provided for the perceiver through the ability to simulate the motor, affective, and mental states of an observed other, thereby facilitating a form of indirect 'access' to the internal state of the observed agent (Gallese, 2001).

Similarly, the perception-action model (PAM), proposed by Preston and de Waal (2003), is based on the perception-action hypothesis, which proposes that the perception of behaviours and emotions of others automatically triggers, within the observer, internal representations and activation of neural mechanisms responsible for those emotions and behaviours (Prinz, 1992). This particular system prompts the perceiver to emotionally converge or resonate with another, as a direct result of activation of motor representations that stem from the observed target, and subsequent changes in the autonomic nervous system (Preston & de Waal, 2003). It is theorised that this occurs due to the perceiver's tendency to generate an identical (or near-identical) psychophysiological emotional state to an observed

target as a result of similar biological programming of their respective autonomic nervous systems (Basch, 1983). That is, a 'physiological linkage' between the perceiver and observed agent occurs due to a commonality of emotional experience, and reliable 'mapping' of physiological patterns of activity onto motor and emotional states (Basch, 1983; Levenson & Ruef, 1992). This model is consistent with the common-coding theory, which states that actions are 'coded' by their observable effects, and share a common-code of internal representations within the brain (Preston & de Waal, 2003; Prinz, 1992). Perceivers translate and use their own representations of behaviours and associated emotions to feel and understand the mental state of the observed other (Preston, 2007). MNS properties are consistent with the PAM and common-coding theory, whereby congruent internal representations are activated during the observation of the actions of others via a simulation mechanism, which facilitates the observer's ability to empathise with, and ultimately understand, the emotional and mental world of the observed other.

Researchers have also proposed a predictive coding account for the MNS in understanding others. According to this account, the brain acts as a 'hypothesis tester', constantly receiving and updating sensory input from the perceived environment, integrating information with stored knowledge gained through prior experience, and minimizing prediction error (i.e., the difference between predicted and actual sensorimotor outcomes; Friston, 2010; Urgen & Miller, 2015). In other words, the ability to infer another's intentions through observation of actions occurs via internal one-to-one mapping between sensory input and associated intentions within the perceiver. The predictive coding account (and the notion of minimising prediction error) relies heavily on shared representations of one's own experience and that of another (i.e., there are assumed commonalities between the stored understandings and experiences of motor, emotional, and mental states between the observer and that of the observed agent; Brown & Brüne, 2012). Due to its action-observation properties, the MNS fits well within this account, providing an implicit link between the

observer and the observed other. This interpersonal connection of shared understanding and experience is thought to lead to, or facilitate, empathic experiences (Brown & Brüne, 2012; Decety & Ickes, 2011). Based on this theory of predictive coding, the mental states of others are predicted based on the interaction between sensory inputs (i.e., observed behaviour and observed motor/physical information) and prior knowledge or experience (i.e., stored internal representations through personal social experiences; Brown & Brüne, 2012).

A further account of the MNS in understanding and predicting mental states is based on Hebbian associative learning theory (Heyes, 2012; Heyes & Ray, 2000). Here, mirror neurons are thought to emerge following the formation of learned associations between visuomotor observations experienced within the self and that observed in others (Catmur, Walsh, & Heyes, 2009; Brown & Brüne, 2012). Associative learning begins early on in life, and propels infants onto a trajectory of learning to understand the minds of others (Meltzoff & Decety, 2003). Throughout early life, bidirectional associations between mental states and behaviours develop through the experience of motor behaviours (such as facial expressions) and subjective inner feelings (Meltzoff & Decety, 2003). That is, through day-to-day practices, infants develop a detailed and comprehensive 'map' of associations between their own physical actions and their corresponding mental and emotional experiences. This 'map' informs the ability to make inferences about the internal experiences of others through observing actions and projecting the same mental experiences of the self onto others, on the basis of the observed behaviour (Meltzoff & Decety, 2003). This 'map' is continuously developed and refined throughout life and used to predict intentions and behaviours of others, aiding empathic experiences and social interactions.

Collectively, and individually, these theories imply that the MNS may play a role in empathy and, more broadly, social cognition. When evaluating these theoretical accounts in attempting to explain how action-observation of motor behaviour provided by mirror neurons might contribute to the experience of empathy, a number of common and consistent

assumptions between theories emerge. These can be cohered into a series of important suppositions: (1) there are important bidirectional associations that exist between motor behaviours and corresponding emotional and cognitive experiences; (2) these associations are developed throughout life into a comprehensive 'map', which is continuously refined through social interactions and interpersonal experiences; (3) this 'map' provides an implicit link between the perceiver and observer through modelling the internal experiences of others via sensory inputs (i.e., perceived motor actions) or observed behaviours; (4) this model is enabled via *simulation* mechanisms provided by the action-observation properties of the MNS; and (5) these acquired 'maps' or models provide necessary elements for empathising with others and allows predictions to be made regarding the other's mental states.

The summary model for empathy presented in *Figure 1* is consistent with these theoretical frameworks. Specifically, this review considers empathy to involve multiple processes, including motor mimicry or imitation (motor empathy), subsequent emotional and physiological resonance (emotional empathy), and the ability to understand the cognitive and affective states of another (cognitive empathy). Conceptualisations of empathy and theoretical models of MNS both consider *simulation* as an important and necessary mechanism, suggesting the two may be compatible. It is important to note, however, that the direction of this presumed relationship remains unclear. Although some 'virtual lesion' studies using TMS have assessed the role of particular mirror neuron regions, such as the IPL, in various cognitive abilities (e.g., Uddin, Molnar-Szakacs, Zaidel, & Iacoboni, 2006), current research is lacking in more comprehensive examinations of causal links in order to determine the extent of MNS involvement in empathy. The current literature examining the *relationship* remains controversial for a number of reasons. First, due to inconsistencies and variations in conceptualisations of empathy, where studies adopt different definitions and measurements of the same construct. Second, methodological uncertainties and practical limitations of techniques used to acquire mirror neuron activity. Third, the correlational

nature of evidence for mirror neuron involvement and social cognitive processes important for empathy, resulting in uncertainty regarding whether the MNS is *causally* involved in empathy. Fourth, due to the typical practices in neuroscience research of obtaining and relying on small and underpowered samples (Button et al., 2013). Lastly, the current theories, while plausible, do not have robust validation through empirical studies.

1.5. Purpose of the Current Review

Although there are a number of compelling theories regarding the potential role of the MNS in empathy, it remains unclear whether the MNS is necessary or sufficient for empathy to occur, and, to our knowledge, empirical studies investigating this relationship have not been reviewed systematically. The current study aims to provide an overview of existing empirical studies investigating the relationship between key sub-components of empathy (according to the summary model adopted here) and the MNS, and to evaluate evidence concerning this relationship. Additionally, using meta-analytic procedures, this review aims to provide statistical estimates regarding the magnitude of this relationship. Of particular interest, we also aim to examine potential relationship with the MNS and each component of the empathy model used here, in order to examine the individual contributions of each process. A further goal of this review is to examine methodological differences in the measurement of both empathy and the MNS, in order to explore the potential effects of these variations. Finally, we aim to highlight the current gaps and methodological limitations within the literature, and provide recommendations for future studies.

2. Methods

2.1. Study Design

Studies included in this systematic review and meta-analysis must have examined (a) *at least* one domain of empathy (i.e., motor, emotional, or cognitive) and (b) elicited and

neurofunctionally measured putative activity of the mirror neuron system in a healthy, typically-developing population.

Databases were searched in August 2016, using the following search terms limited to academic journals, studies published in English, and human-only studies: [(Mirror neuron* OR action observation OR mu suppression OR interpersonal motor resonance OR fronto parietal action observation network OR parieto frontal action observation network OR FPAON OR PFAON OR biological motion) AND (Mimicry OR imitation OR motor empathy OR facial EMG) AND (Emotional resonance OR emotional contagion OR emotional empathy OR affective empathy OR physiological synchron* OR emotional synchron*) AND (Perspective taking OR mentalising OR mentalizing OR theory of mind OR TOM OR cognitive empathy OR facial affect recognition OR facial emotion recognition OR emotion* recognition)]. See supplementary information for a summary. Databases searched PSYCINFO (via Ebscohost), MEDLINE (via Ebscohost), Scopus, Web of Science, EMBASE, PubMed, Science Direct, and CINAHL (via Ebscohost).

2.2. Study Inclusion Criteria

To be included in the systematic review and meta-analysis, studies were required to be: (1) conducted on human subjects; (2) published in English; (3) not in the form of a thesis, dissertation, conference abstract, poster, opinion piece or non-empirical work; (4) not published as review or book chapter; (5) published after 1995/6, which is when mirror neurons were first reported in humans (Rizzolatti, Fadiga, Matelli, et al., 1996); (6) examining at least one component of empathy (motor, emotional, or cognitive); (7) including an observed outcome variable of the component assessed (i.e., not passively observing stimuli), where participants were required to respond to a stimulus or task; (8) including behavioural outcomes completed by the individual rather than assessed by another (e.g., a family member).

There were also important exclusion criteria: (1) reaction time was not a variable of interest in this review, as speed of response is beyond the scope of this study, and was not considered as important as accuracy in reflecting empathic abilities; (2) regarding motor empathy studies, measures of imitation and/or mimicry must have used *emotionally expressive body language* (e.g., facial expression), as opposed to imitation of non-expressive movements (e.g., finger tapping), as non-expressive movement would not be considered 'motor empathy'; (3) studies examining "action understanding" were not considered "cognitive empathy" (because they do not require mental state inference), and were therefore excluded, unless the task involved implicit higher-order perspective taking abilities, as opposed to the understanding simple actions; (4) with respect to the measurement of MNS activity, studies must have included a functionally acquired brain index of MNS activity, and we excluded studies that solely relied on behavioural measures of the MNS and/or used only structural scans were excluded; (5) studies presenting static stimuli were only included if the study intended to evoke MNS activity, as including any study presenting static images, without the intention of inducing a mirror neuron response would result in the inclusion of studies beyond the scope of this review; (6) the current study aimed to focus on the VM-MNS, studies eliciting the extended MNS (i.e., studies where changes in MNS activity were not solely driven by the visuo-motor MNS, such as pain studies) were excluded; (7) studies examining clinical populations without healthy controls were also excluded, as the focus of the current review is in healthy populations; (8) studies that conducted a correlation analysis on a pooled clinical and control sample, as opposed to running separate analyses per subsample, were not included and finally; (9) studies were excluded from the meta-analysis if they did not correlate MNS activity with at least one domain of empathy. A summary of these criteria is presented in Table 1.

Table 1. A Summary of Exclusion and Inclusion Criteria for the Current Review

Empathy Component	**Definition and Conceptualisation**	**Measurement**
Motor*	The automatic mimicry and synchronization of body language, such as facial expressions, during the observation of another	Detection of changes in facial muscles or body language in response to that of another using: • Facial electromyography (EMG) or; • Facial Action Coding System (FACS) coded by a professional
Emotional**	The ability to immediately detect and physiologically and/or emotionally resonate with the emotional state of an observed other	Physiological and/or emotional assessment of changes in arousal following emotional display of another, such as: • Cardiovascular changes (HR, BPM, Respiration), electrodermal changes (e.g., GSR) • Self-reported changes
Cognitive***	The ability to identify and understand the mental state of another. This includes comprehending the thoughts (cognitive) and feelings (affective state) other another person.	Assessment of the ability to recognise the internal state (thoughts and emotions) of an observed other using: • Assessments of accuracy and/or RT of emotion/mental state recognition through observing faces or interactions, social inference tasks • Self-report measures
Excluded if	Study does not include: (a) at least one measure of empathy; (b) an outcome measure of empathy (i.e., not passively observing stimuli); and (c) measures taken from the individual (i.e., not assessed solely by a friend/family member).	
Mirror Neuron Activity	**Definition**	**Measurement**
	Specialised cortical cells that fire during both the execution of an action and during the observation of a similar action, found primarily in the IFG and IPL.	Neuro-techniques assessing changes in cortical activity during action observation (AO) via: • BOLD response – fMRI • Motor evoked potentials – TMS • Mu suppression (7.5 – 12.5 Hz) – EEG &/or MEG AO stimuli include: • Biological motion; static and dynamic presentations of faces, hands, fingers (human or animated/naturalistic)
Excluded if	Study does not include: (a) a brain index of mirror neuron activity (MNA); (b) a functionally acquired indicator of MNA (i.e., assess MNA during presentation of AO stimuli); (c) an accepted method of eliciting MNA (such as those listed above); (d) presentation of static images, *with* the intention of assessing MNA (i.e., studies presenting static stimuli without aiming to examine the MNS will not be included – this does not apply to dynamic stimuli or biological motion due to theoretical reasons) and (e) stimuli used only elicits visuomotor MNS,	
Other important exclusions	(a) studies only examining clinical populations were excluded; (b) studies that only provided correlations of a pooled clinical and healthy sample; (c) studies were excluded if they did not correlate MNS activity with at least one domain of empathy	

Note. Motor* = synonymous with: Motor resonance/Simulation, Motor mimicry, and Imitation; Emotional** = synonymous with Affective empathy, Emotional/affective resonance Emotional/affective contagion, and Emotional/affective synchrony; Cognitive***= synonymous with Theory of Mind (ToM), Perspective taking, and Mentalising. IFG = inferior frontal gyrus; IPL = inferior parietal lobule.

2.3. Data Extraction

A random sample at both full-text review and data extraction stages were screened by a second reviewer to ensure consistencies between extractors. Inter-rater reliability was determined using prevalence-adjusted bias-adjusted kappa (PABAK; Byrt, Bishop, & Carlin, 1993). Data extracted from the final sample of studies included basic demographic information, such as ethnic diversity of sample, sample size, sample type (healthy or clinical and healthy), age (*M*, *SD*, range), sex ratio, years of education (*M*, *SD*), and whether the sample were right-handed (Yes or No). Next, we extracted data pertaining to the method used to functionally acquire a neural index of MNS activity in terms of the neuroscience technique used (fMRI, TMS or EEG), as well as the stimulus presented to elicit a response (e.g., of observation of facial expressions). Related to this, we also extracted information regarding differences in the *property* or *type* of stimulus used, such as whether the stimulus was (1) dynamic or static and (2) stylised (i.e., cartoons/avatars) or natural (i.e., human faces).

Information regarding experimental design, such as types of control conditions/baseline, as well as passive observation and/or active execution, were also collected. Regarding measurement of empathy, we recorded the component of empathy measured, the type of measurement used to assess empathy, whether this was self-report or experimental, the type of outcome the measure used would produce (e.g., accuracy rates or emotion ratings), and the final scores acquired from these measures. We also included direct quotes or summaries of the conceptualisation and/or definition of empathy used by each study to highlight the variability of the description and understanding of this construct between studies. Where studies did not differentiate between cognitive and emotional empathy (that is, these were measured in unison using total scores of empathy questionnaires; see introduction subsection ‘Conceptualisations of Empathy and its Measurement’), we categorised these as ‘combined cognitive and emotional empathy’. Finally, concerning

unpublished data, authors who did not report all necessary data in the published manuscript were contacted and results were requested.

2.4. Meta-Analytic Procedures

To be included in the meta-analysis, studies must have correlated at least one outcome measure of empathy (motor, emotional, or cognitive) with a functionally acquired neural index of the mirror neuron system in a healthy, typically-developing human sample. Additionally, at least two studies are required for a meta-analysis to be run. It should be noted that studies using fMRI were meta-analysed according to the specific region of the MNS (i.e., inferior frontal gyrus or inferior parietal lobule), as these regions have been theoretically linked to different processes and outcomes (Carr et al., 2003; Iacoboni, 2009). Additionally, concerning the inclusion of fMRI studies, we attempted to account for different methodological approaches (i.e., whole-brain or a-priori ROI-based approaches and peak-voxel approaches) in order to account for potential artificial inflation due to p-value selection biases. However, during the extraction process, we found high variability in approaches across studies that made it difficult to categorise methodology. Where studies did not provide a correlation coefficient as an effect size, we used equations outlined in (Friedman, 1982) to convert available test statistics into a correlation coefficient. Additionally, we used Cohen's (1992) guidelines for interpreting effect sizes.

All meta-analytic computations were conducted in R (Team, 2016). In this meta-analysis, most included studies violated the assumption of independence, with studies reporting multiple effects (*r*) from within the same sample, likely owing to the multi-faceted nature of empathy, as well as the numerous methods of measuring MNS activity. To include all available data, we utilised the meta-analytic technique of robust variance estimation using the Robumeta package in R (v3.3.2; Fisher, Tipton, Zhipeng, & Fisher, 2017), a random-effects meta-regression that statistically accounts for multiple dependent effects (Hedges, Tipton, & Johnson, 2010; Tanner-Smith, Tipton, & Polanin, 2016). This technique supports

the use of clustered data via robust estimations of effect size weights and standard errors to account for within sample correlation . This method allows for multiple outcomes within a study, and eradicates the need to select only one effect, or calculating an average of all effects per study (Hedges et al., 2010; Tanner-Smith et al., 2016; Tanner-Smith & Tipton, 2014; Tipton, 2013). For studies that only consisted of one effect, a standard meta-analysis was run.

Finally, heterogeneity was assessed using the calculation of I^2 (Higgins, Thompson, Deeks, & Altman, 2003), which evaluates the degree of inconsistency across studies and ranges from 0 to 100%, with high scores indicating that there is a high proportion of the total variability due to between study variation. The I^2 statistic for each analysis is presented in table 5. Note that 95% confidence intervals for the I^2 are not yet available for robust variance meta-analysis.

2.5. Moderation analyses.

Meta-regression was used to evaluate the potential moderation effect of methodological differences implemented between studies on observed outcomes. Where possible (that is, where at least two or more studies were available for each level of the comparison), we examined whether the meta-analytic correlation effect was moderated by: (1) mode of empathy measurement, assessing differences between self-report and experimental measures; (2) stimulus properties used for eliciting a mirror neuron response, comparing the use of static and dynamic stimuli; and (3) the type of those images used, evaluating potential differences between natural (e.g., human faces) and stylized (e.g., cartoons/avatars) images. If a moderation analysis could not be run due to too few effect sizes in one of the two comparison categories of the moderator, a sensitivity analysis was conducted for only one of the comparison categories if that comprised at least two independent studies for analysis. This sensitivity analysis therefore allowed comparison of the meta-analytic obtained in the subgroup relative to the overall meta-analytic effect obtained from all studies in the respective meta-analysis.

3. Results

3.1. Narrative Review of Included Articles

Figure 2 presents the PRISMA flow diagram of studies identified in the search. Of the initial 7094 articles found, 3688 represented unique articles. After title and abstract screening, 845 studies remained for full text review. The remaining 845 full-text articles were assessed for eligibility using the criteria outlined in detail in table 1. A random sample of 10% of the full-text articles (n = 84) were screened by a second reviewer (P.G.E.). The inter-rater reliability between the two reviewers was strong (PABAK = 0.810; Byrt et al., 1993). A second random sample of 37 of the full-text articles during data extraction (100% of studies where an effect was provided, n = 27; and 50% of studies where no effect was provided, n = 10) were screened by a second reviewer (P.H.D.). The inter-rater reliability between the two reviewers was very strong (*PABAK* = 0.919; Byrt et al., 1993). Any inconsistencies or disagreements were resolved by discussion. A detailed summary of all included studies is provided in tables 2, 3, and 4. Finally, a total of 41 authors from the included studies were contacted to request unreported and unpublished data. Thirty correlations were provided by seven of the contacted authors, and these were included in the final meta-analysis.

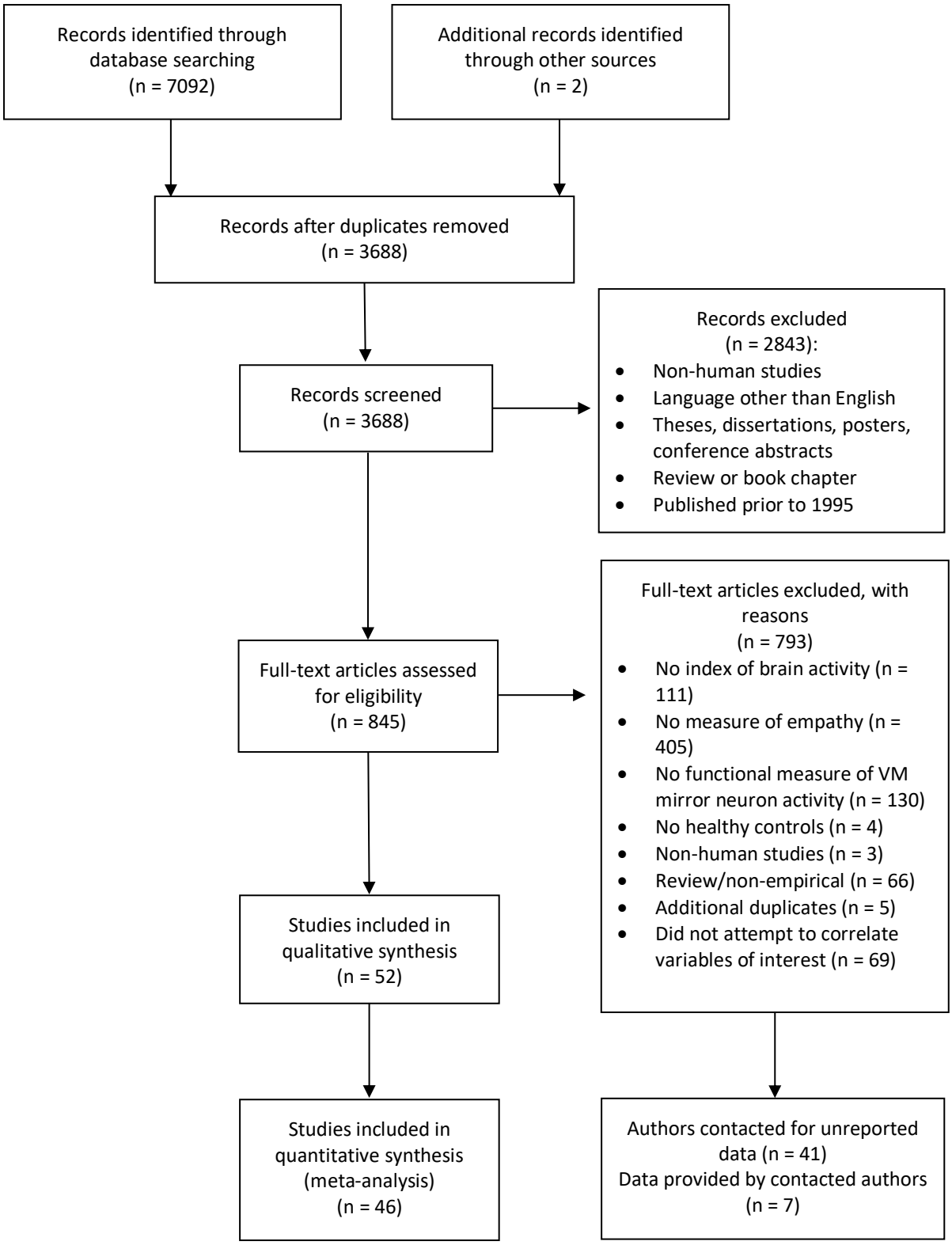


Figure 2. PRISMA flow chart

Table 2. Characteristics of Included Studies

Authors (year)	Sample Type	Sample Size	Age *M* (*SD*)	Age (years) *Range*	Sex (*M/F*)	Education M (*SD*)
Schulte-Ruther et al (2014)	Clinical (*ASD*) and Healthy	27 (*54 including clinical group*)	18.22 (4.41)	12 - 26	27/0	N/A
Ihme et al (2014)	Healthy†	48	24.00 (3.00)	18 - 29	25/23	N/A
Lenzi et al (2013)	Healthy†	23 (*'secure' subjects, n = 11 and 'dismissing' subjects, n= 12*)	23.45 (N/A)	20 - 28	0/23	N/A
Woodruff & Shelley (2013)	Healthy†	23	19.00 (1.44)	18-23	7/16	N/A
Libero et al (2014)	Clinical (*ASD*) and Healthy	23	21.38 (1.44)	13-36	20/3	N/A
Ferri et al (2014)	Clinical (*Schizophrenia*) and Healthy†	22	28.00 (3.77)	N/A	12/10	N/A
Babiloni et al (2012)	Healthy	22 (*professional saxophonists [S], n =12, & non-musicians [NM], n =10*)	*S*: 35.90 (3.54); *NM*: 52.20 (2.90)	*S*: 21-50; *NM*: 34-64	22/0	N/A
Cooper et al (2012)	Healthy	19	22.00 (5.00)	N/A	5/14	N/A
Milston et al (2013)	Healthy†	26	21.73 (8.24)	N/A	6/20	N/A
Anders et al (2012)	Clinical (*Parkinson's Disease*) and Healthy	8	*Males*: 44.25 (N/A), *Females*: 46.75 (N/A)	35-55	4/4	N/A
Jabbi et al (2015)	Healthy†	40	30.54 (N/A)	N/A	33/17	N/A
	Healthy (*sub sample of above*)	21	33.09 (N/A)	N/A	14/7	N/A
Hoenen et al (2013)	Healthy†	28	29.00 (9.00)	18-62	12/16	N/A

Lepage et al (2010)	Healthy†	23	22.50 (2.21)	N/A	7/5	N/A
Moore et al (2012)	Healthy†	22	N/A	N/A	11/11	N/A
Bernier et al (2007)	Clinical (*ASD*) and Healthy[a]	15	23.60 (4.90)	19.4-37.5	15/0	N/A
Lepage et al (2014)	Clinical (*Turner's Syndrome*) and Healthy	15	26.10 (6.80)	N/A	0/15	N/A
Perry et al (2010)	Healthy†	24 *(analysed*)	24.30 (N/A)	19-28	11/15	N/A
Enticott et al (2008)	Healthy	20	28.70 (7.30)	N/A	8/12	N/A
Enticott et al (2011)	Healthy†	37	26.51 (5.64)	19-38	19/18	N/A
Jola et al (2012)	Healthy[b] *(subgroups = experienced ballet dancers [EB]; experienced Indian dancers [EI]; novice [Nov])*	29 (*EB, n = 12; EI, n = 8; Nov, n = 9*)	*EB*: 52.4 (13.8), *EI*: 51.6 (17.5), *Nov*: 21.0 (10.0)	20-72	10/19	N/A
Andrews et al (2015)	Clinical (*Schizophrenia*) and Healthy	19	37.84 (13.07)	24-64	N/A	15.74 (2.28)
Mehta et al (2014)	Clinical (*Schizophrenia*) and Healthy	45	30.68 (9.57)	N/A	23/22	13.13 (3.50)
Brown et al (2016)	Clinical (*Schizophrenia*) and Healthy†	17	38.29 (12.10)	N/A	9/8	N/A
Bernier et al (2013)	Clinical (*ASD*) and Healthy	19 (*analysed*)	6.40 (1.30)	N/A	17/2	N/A
Silas et al (2010)	Healthy†	33 (*analysed*)	25.70 (N/A)	18-39	16/17	N/A

Makhin et al (2015)	Healthy†	44 (*completed behavioural testing, n = 28*)	N/A	18-24	8/37 (*7/21*)	N/A
Woodruff et al (2016)	Healthy†	30	19.76 (2.62)	18-30	13/17	N/A
Moore et al (2016)	Healthy	19	21.25 (N/A)	18-25	7/12	N/A
Schulete-Ruether et al (2011)	Clinical (*ASD*) and Healthy	14 (*analysed*)	25.05 (6.69)	19-46	18/0	N/A
Castelli et al (2010)	Healthy†	24 (*'young', n = 12; 'old', n = 12*)	*Young*: 25.20 (3.50); *Old*: 65.20 (5.70)	*Young*: 21 - 30; *Old*: 60 – 78	*Young*: 2/10; *Old*: 4/8	N/A
Mazzola et al (2013)	Healthy	23	29.00 (7.50)	N/A	11/12	N/A
Hadjikhani et al (2014)	Clinical (*ASD*) and Healthy	31 *(analysed)*	22.50 (7.50)	N/A	28/3	N/A
Likowski et al (2012)	Healthy†	20	23.50 (3.05)	20 -30	0/20	N/A
Schulte-Ruther et al (2008)	Healthy†	26	*Males*: 24.40 (3.00); *Females*: 24.80 (3.70	N/A	12/14	N/A
Kaplan et al (2006)	Healthy†	22	26.00 (6.00)	N/A	9/13	N/A
Moriguchi et al (2009)	Healthy† (*divided into high vs low Alexithymia*)	37	20.40 (0.92)	N/A	7/30	N/A
Mier et al (2010)	Healthy†	40	25.25 (3.52)	19-32	20/20	N/A
Lee et al (2014)	Clinical (*Schizophrenia*) and Healthy†	16	36.80 (6.30)	N/A	10/6	14.8 (2.8)
Hooker et al (2008)	Healthy	20	21.00 (N/A)	19-26	9/11	N/A
Schulte-Ruther et al (2007)	Healthy†	26	24.60 (3.40)	N/A	12/14	N/A

McCormick et al (2012)	Clinical (*Schizophrenia*) and Healthy	16	36.60 (9.70)	N/A	14/2	15.4 (1.4)
Hooker et al (2010)	Healthy	15	21.00 (N/A)	18-25	8/7	N/A
Zaki et al (2009)	Healthy†	21	19.10 (N/A)	N/A	10/11	N/A
Greimel et al (2010)	Clinical (*ASD*) and Healthy	24 (*Adolescent Controls, n = 15*; *Fathers, n = 9)*	*Adolescents*: 15.00 (1.40); *Fathers*: 43.90 (5.10)	N/A	24/0	N/A
Kana & Travers (2012)	Healthy†	26	21.00 (N/A)	18.5 - 35.8	12/14	N/A
Mier et al (2014)	Clinical (*Psychopathy*) and Healthy	18	44.0 0(10.35)	N/A	18/0	10.11 (1.13)
Horan et al (2014)	Clinical (*Schizophrenia*) and Healthy[c]	23	46.70 (6.90)	N/A	16/7	14.9 (1.6)
Braadbart et al (2014)	Healthy†	20	*Males*: 28.90 (8.26): *Females*: 23.80 (3.11)	N/A	10/10	N/A
Pichon et al (2009)	Healthy†	16	*Males*: 23.50 (2.60; *Females*: 25.60 (8.0)	N/A	8 /8	N/A
Alaerts et al (2014)	Clinical (*ASD*) and Healthy	15	23.30 (2.90)	N/A	15/0	N/A

Flourney et al (2016)	Healthy[d]	*Wave 1, n* = 56; *Wave 2*, *n* = 56	*Wave 1:* 10.10 (0.31); *Wave 2:* 13.10 (0.31)	N/A N/A	26/30 27/30	N/A N/A
Woodruff et al., 2011	Healthy	39	N/A	N/A	16/23	N/A

Note. [†]Sample consisted of right handed participants. [a]Ethnicity recorded: White (86.6%); 'Other'(13.3%). [b]Ethnicity recorded: Caucasian U.K. (65.5%), Caucasian mainland Europe (13.8%), Black or mixed skin colour (3.5%), India (17.2%). [c]Ethnicity recorded: White (73.9%), African American (26.1%), Hispanic (4.3%). [d]Ethnicity recorded: White (47%) Hispanic/Latino (12%), Black/African American (4%), American Indian/Native Hispanic/Latino (2%), Asian or pacific Islander (2%), Multiracial (30%), 'Other' (4%)

Table 3. Summary of Conceptualisations of Empathy and Methodological Approach Across Studies

Authors (year)	Conceptualization of Empathy	Task used to measure empathy component/s (experimental)	Task used to measure empathy component/s (self-report)	Empathy outcome variable
Schulte-Ruther et al (2014)	"…the result of psychological inferences about other persons' mental and emotional states, allowing for socially appropriate response. Empathy is a multidimensional construct entailing emotional aspects (shared affect and emotional responses) as well as cognitive aspects (such as perspective-taking, self-other distinction, reflection about other people's mental states, and explicit self-assessment of own evoked emotions)". ***C, E, C&E***	Assessment of the ability to judge emotional state of another	Self-reported emotions in response to stimuli (facial expressions); EQ; Bryant Index of Empathy for children/adolescents	EQ scores; Bryant Index of Empathy scores; Mean RTs and % of correct attribution of emotional states (of stimulus face during other task) & congruent responses (i.e., same emotion displayed as stimulus expression presented) during self-task
Ihme et al (2014)	*Discussed in the context of facial expression and recognition:* *"*Understanding the emotional expression of another person is thought to require mimicry or simulation of others' facial expressions". ***C, E, C&E***	Assessment of ability to label facial emotion of a target face (happy vs angry vs fearful vs neutral); and the Toronto Structured Interview for Alexithymia (German version - administered by trained interviewer) - focused on subscale '*difficulties describing feelings'*	TAS-20 (German version) - focused on subscale "*difficulties describing feelings*"	TAS-20 scores & facial labelling accuracy

Lenzi et al (2013)	"Empathy provides a comprehensive account of intersubjective intercourses, enabling individuals to establish a meaningful connection with the others' emotions". *These proposed components include:* "affective sharing between the self and the other which may be conceptualized as the ability to detect and resonate with the immediate affective state of another" *and* "…self-other awareness, without which the only affective sharing would lead to the phenomenon of emotional contagion, that is, the 'total identification without discrimination between one's on feelings and those of the other'" *and finally,* "mental flexibility to adopt the subjective point of view of the other". ***C, E***	'*Relative Functioning*' (ability to describe another's mental state [feelings, wishes, thoughts, intentions, and desires] as measured by the Adult Attachment Interview (AAI; scored by certified coders)	TAS-20	TAS-20 scores
Woodruff & Shelley (2013)	*No explicit definitions provided, however, discussed simulation mechanisms:* "…whereby the observed motor actions of another person are rapidly converted into sensorimotor signals within the observer's brain, creating a neural simulation of the conspecific's intentions". ***C&E***	N/A	EQ & IRI	EQ & IRI scores
Libero et al (2014)	Discussed empathy in terms of body expression, which was described as a conveyer of emotional states of others, and assists in inferring others' feelings and intentions. ***C, E, C&E***	Emotion attribution accuracy of body posture presented & RMET	EQ	Accuracy of emotion attribution RMET accuracy, & EQ scores
Ferri et al (2014)	*No explicit definitions provided: mainly discussed empathy as an affected component of social cognition.* ***C&E***	N/A	EQ	EQ scores
Babiloni et al (2012)	"The definition of empathy encompasses at least two main dimensions. 'Emotional' empathy defined as the immediate affective 'contagion' of feelings between the observer and observed person. Instead, 'cognitive' empathy is based on thinking, and can be defined as the capacity to adopt the psychological point of view of another person thanks to a perspective-taking ability". ***C&E***	N/A	EQ	EQ scores
Cooper et al (2012)	*No explicit definitions provided. Discussed empathy in the context of contagious yawning via mirroring or simulation mechanisms* ***C&E***	N/A	IRI	IRI scores

Milston et al (2013)	*Adopted the perception-action model for empathy:* …perception of another's state will automatically activate representations of that state in an individual, which in turn generates autonomic and somatic responses unless consciously suppressed". ***C&E***	N/A	IRI - Empathic Concern & Perspective Taking subscales only	IRI scores
Anders et al (2012)	*No explicit definitions provided: discussed motor impairments and associated cognitive and affective dysfunctions in patients with Parkinson's disease.* ***C, E***	Accuracy of emotion attribution performance on the 'Facial Expression of Emotions: Stimuli and Test (FEEST)' consisting of surprise-fear-sadness-disgust-anger-happiness (conducted prior to imaging)	'*Joy'* rating in response to stimuli in scanner (*1= not at all, 7 = very intense*)	Performance accuracy (FEEST) & SR joy rating
Jabbi et al (2015)	*No explicit definitions provided: discussed the importance of dynamically changing facial muscles as providers of understanding emotional states* ***C, E***	Post-scan accuracy of emotion recognition of the same dynamic task viewed during imaging	N/A	RT &accuracy of emotion recognition of faces
Hoenen et al (2013)	"… a process leading to shared feelings. This process can be divided into a cognitive-evaluative and an affective-perceptual component"; "cognitive empathy is linked to ToM, the knowledge about the intentions and thoughts of others, and includes the process of putting oneself in the perspective of others and imagining their feelings. Affective empathy is defined as automatic processes leading to the re-experience of the feelings of someone else (e.g., emotional contagion)". ***C, E, C&E***	Self-reported perspective taking & empathic concern of the actor	Rated '*how emotional*' the narration of an actor was & the degree of affection experienced while watching the video	PT & EC ratings
Lepage et al (2010)	*Considered a motor theory of empathy*: "…a basic action mapping system would contribute to higher order cognitive processes, such as emotional sharing and empathy". ***C&E***	N/A	EQ	EQ scores

Moore et al (2012)	*Discusses simulation theories for the processing of emotional*: "One of these is facial mimicry, activation of the facial muscles used to produce a given emotion expression in response to seeing the expression on another face. A second type of facial simulation is facial mirroring, activation of neural substrates for expression production that is not for the sake of mimicry but rather for producing an "offline" simulation of what is observed. Facial mirroring, which refers primarily to the mirroring of the action of generating an emotional facial expression, is believed to be important for making inferences about how others are feeling". ***E***	N/A	Balanced emotional empathy scale (BEES) & mood rating during empathy task (*1-5 Likert scale*)	BEES scores & Mood rating
Bernier et al (2007)	*No explicit definitions provided.* ***M***	Imitation ability/accuracy (coded by trained individual using the Mature Imitation Task Manual)	N/A	Imitation ability/accuracy
Lepage et al (2014)	*No explicit definitions provided.* ***C&E***	EQ	EQ	EQ scores
Perry et al (2010)	*Discusses the social-cognitive simulation theory:* "…the information on which ToM skills are based is not sensory but rather motor in nature"; "Simulating the actions performed by others and associating the simulated action with motor representations of our own internal states, motivations, and intentions is hypothesized to be a general mechanism whereby we are able to generate knowledge of other minds". ***C&E***	N/A	IRI & EQ	IRI & EQ scores
Enticott et al (2008)	*Discusses facial emotion processing* "the ability to identify emotion through the observation of facial expression" *and simulation mechanisms important for allowing internal modelling of another's facial expression and assisting in developing understanding of another's mental and affective states*. ***C***	Visual discrimination task (facial emotion) & emotion recognition task	N/A	Emotion discrimination & recognition accuracy of performance

Enticott et al (2011)	*No explicit definitions provided: however, described "embodied simulation of other's mental states as facilitator to understanding the thoughts, beliefs, and emotions, of others.* ***C&E***	N/A	EQ	EQ scores
Jola et al (2012)	*Discusses the direct-matching hypothesis:* "…to understand the meaning of actions by means of internal simulation". ***C&E***	N/A	IRI	IRI scores
Andrews et al (2015)	*Discusses ToM* "...the ability to predict others' thoughts and feelings" *and recognising emotions from faces.* ***C***	1- *NimStim Face Task*: showing '*happy*', '*angry*', '*fearful*', '*sad*', '*disgusted*' or '*surprised*' faces and 2- *Cognitive and Affective Mental Inference Task*: ToM task	N/A	(1) Overall accuracy of performance score where participants selected which of the 6 emotions fit the expression presented (2) accuracy & RT (*ms*) of performance
Mehta et al (2014)	*Discusses the simulation and other theories, such as the 'neural exploitation hypothesis':* "This theory suggests that we reuse our own mental states represented with a bodily format in functionally attributing them to others" *and the 'social projection theory':* "suggests that knowledge of oneself is used as a platform from which we understand others. ***C***	ToM scores using the social cognition rating tools in Indian settings (SOCRATIS) - Sallyanne task & smarties task (1st order), 2nd order false belief picture stories and missing cookies, 2 metaphor-irony stories and 10 faux pas recognition stories	N/A	Individual & total scores on SOCRATIS

Brown et al (2016)	*No explicit definitions provided.* ***C, C&E***	The Unexpected Outcomes Test (assessment of affective reasoning and understanding of others mental states and emotions)	IRI	IRI scores & 'unexpected outcomes' test performance
Bernier et al (2013)	*No explicit definitions provided.* ***M***	Imitation ability/accuracy (coded using the Mature Imitation Task Manual)	N/A	Accuracy of imitation (hand, finger, & facial)
Silas et al (2010)	*No explicit definitions provided.* ***C&E***	N/A	EQ & IRI	EQ & IRI scores
Makhin et al (2015)	*No explicit definitions provided.* ***C&E***	N/A	Questionnaire of Emotional Empathy (QMEE)	QMEE Scores
Woodruff et al (2016)	"Of all the emotions one might feel, only a portion of them reflect one's own emotional state. Some emotions one feels are mirror reflections of a conspecific's emotions — one experiences them, not because they are her own but because she observed the conspecific expressing them". *Also discusses perspective taking:* "The ability to discriminate one's intentions from another's is critical to perspective-taking" *and self-other distinctions:* "…discriminating feelings one experiences because it reflects one's own emotional state from those feelings one experiences because one observed another expressing them". ***C, E, C&E***	Facial emotion attribution	Self-reported internal feelings in response to facial emotions, IRI, & EQ	Facial emotion attributional accuracy & self-reported feelings
Moore et al (2016)	*"*...action observation allows humans to 'simulate' the action in their own minds". ***C***	Facial emotion attribution	N/A	Facial emotion attributional accuracy
Schulete-Ruether et al (2011)	"…the result of psychological inferences about other persons' mental and emotional states allowing for socially appropriate emotional responses. The ability to empathize entails both emotional and cognitive components". ***E, C&E***	Subjects were asked to empathise with the facial expressions presented, and were required to judge the emotional state of the face (other)	EQ & subjects were asked to empathise with the facial expressions presented, & were required to report their emotional experience elicited by the presented face (self)	EQ scores & emotion attribution accuracy & self-reported emotion response

Castelli et al (2010)	*Discusses the ability to predict, understand, and explain behaviour and ToM* "…the capacity to understand one's own and other people's mental states (emotions, desires, beliefs) and to refer to them to foresee and explain the behaviour". ***C***	4 ToM tasks: (1) '*Eye direction Detection*'; (2) '*Deceptive box*' (1st order false belief); (3) '*Look-prediction and say prediction*' (2nd order false belief); (4) selection from '*Strange Stories*' & RMET	N/A	Scores for all 4 ToM tasks & RMET scores
Mazzola et al (2013)	*No explicit definitions provided.* ***E***	Rated the emotions ('*joy' vs 'anger'*) of the facial expressions displayed in videos	Intensity of emotional expressions rating	Self-reported ratings of emotions
Hadjikhani et al (2014)	"the ability to form an embodied representation of another's emotional state, while at the same time being aware of the causal mechanism that induced the emotional state in the other" … "empathy is a multicomponent process, consisting mainly of experience sharing and mental state attribution." *Also discusses perception-action coupling*: "embodiment entails the forming of a representation of the other person's feelings, and thereby sharing of their experience". ***C&E***	N/A	EQ	EQ scores
Likowski et al (2012)	*No explicit definitions provided.* ***M***	Facial EMG activity of the zygomatic major (smile) & corrugator supercilii (frown)	N/A	ZM & CS Activity (*M* EMG change from baseline in μV) in response to faces
Schulte-Ruther et al (2008)	"…a complex multidimensional psychological construct which comprises several psychological processes. In its broadest definition empathy can be described as a reaction to observed emotional states in other people which may include (i) cognitive components like perspective taking, mentalising or self–other distinction, and (ii) emotional components such as resonance with the emotions of others…". ***E***	Required to evaluate emotional intensity expressed by stimulus face (other*; range= 1 - 5, 1 = none; 5 = v. strong*)	TAS-20, BEES & Required to concentrate on own feelings elicited (self) & rate emotional intensity (*range = 1 - 5, 1 = none; 5 = v. strong*)	TAS-20, BEES scores & self-reported emotional state & emotion attribution performance

Kaplan et al (2006)	"Activating our own motor representation could allow us also to activate motivations and intentions that are associated with those actions. This ''resonance'' with another individual can also be viewed as a form of empathy…not only do we understand what are the goals of another person, but we experience their intention and therefore their emotion when we watch them behave." ***C&E***	N/A	IRI	IRI scores
Moriguchi et al (2009)	*Discusses Alexithymia in the context of empathy "*…characterized by reduced self- other distinction and immature empathy, such as higher self-oriented personal distress or emotional contagion" … "individuals with ALEX have reduced metalizing capability, cognitive empathy, and perspective-taking ability. ***C, E, C&E***	N/A	IRI & TAS-20	IRI & TAS-20 scores
Mier et al (2010)	*Discusses emotion recognition and ToM:* "…the ability to infer an emotional state of another individual, mainly from acoustic and visual features like vocalization and facial expression…" … "… the ability to attribute mental states, such as beliefs, desires, and intentions to oneself and others" ... "the representation of another person's emotional state and the knowledge about posture and movement direction should culminate in the recognition of intentions. ***C***	Task that evaluated whether a preceding statement (e.g., "this person is going to run away" or "this person is angry") fitted the facial stimulus ('*joy', 'angry', 'fearful', 'disgusted', or 'neutral*')	N/A	Performance on statement & face matching task (RT and accuracy
Lee et al (2014)	*No explicit definitions provided - discusses blunted affect in the context of motor abnormality in schizophrenia.* ***M***	Participants were required to either (1) imitate the facial expression presented or (2) make a facial expression in response to word stimuli such as '*happy*' or '*sad*' Facial expressions were coded two trained coders	N/A	FECS scores in both posed & evoked conditions

Hooker et al (2008)	"... both mentalising and empathy require an understanding of someone else's mental or emotional state, empathy additionally requires sharing the emotional experience of the other person." … "…inferring mental and emotional state from multiple sources, including non-verbal cues, such as facial expressions and gaze direction, as well as knowledge about the other person's perspective and beliefs". ***C, C&E***	Emotion recognition task: required to indicate what emotion a character was experiencing, out of a list of 4 options ('*happy*', '*embarrassed*', '*angry*', '*afraid*'; "true belief" [TB]), & what another character in the scene is feeling ("false belief" [FB]) Emotion Inference task: required to indicate what a character would feel if s/he had "*a full understanding about what is happening in the scene*" for the main ("true belief") and other ("false belief") character in the scene.	IRI	Task performance (emotion recognition and inference tasks) & IRI scores
Schulte-Ruther et al (2007)	"…(i) an intuitive feeling of having something in common with the other person which relies on socially shared emotional experiences; (ii) cognitive mechanisms of perspective-taking; and (iii) the ability to maintain a self–other distinction during interpersonal interaction. Empathic feelings allow for socially appropriate emotional responses, be it a shared or reactive emotional state". ***E***	N/A	BEES & the Empathic Concern Scale (ECS)	BEES & ECS scores

McCormick et al (2012)	"... empathy has both cognitive and affective components and generally refers to the capacity to recognize and share the feelings experienced by another" & "One proposed theory for the ability to understand mental states of others is through simulation theory… for experiencing others' sensory, motor, perceptual, and emotional experiences as if they were one's own". ***C&E***	N/A	IRI	IRI scores
Hooker et al (2010)	"…mentalizing and empathy require an understanding of someone else's mental or emotional state, empathy additionally requires sharing the emotional experience of the other person" … "people simulate the experience of others in order to best understand them". ***C, E, C&E***	<u>Emotion recognition task</u>: subjects required to indicate what emotion a character was feeling, & <u>Emotion inference task:</u> what they *would* feel if he/she had a full understanding of what is happening	IRI	IRI scores, emotion recognition & inference task
Zaki et al (2009)	"'shared representations' (SRs) of experienced and observed affective, sensory, and motor responses allow perceivers to vicariously experience what it is like to be the target of their perception. This common coding between self and other states, in turn, is thought to aid perceivers in understanding targets emotions or intention". ***C***	Videos of autobiographic events of others presented; participants were required to (a) rate how *positive* or *negative* they believed the target felt when the event recalled occurred, and (b) to rate how they believed target felt while talking about the event	N/A	Empathic accuracy (EA) was a reflection of how well the perceivers inferred the target's affect.

Greimel et al (2010)	*No explicit definitions provided, however, discussed impairments in mental state attribution briefly in the context of ASD as well as "empathic deficits".* ***C, E, C&E***	Subjects were asked to empathise with the faces and (1) infer the emotional state from the face ["other" task"]; and (2) judge their own emotional response to the face. Response options were '*sad*', '*neutral*' or '*happy*'	BEES (fathers) & Griffith Empathy Scale (GEM; adolescents)	BEES, GEM, & emotion attribution of self and other scores
Kana & Travers (2012)	"Posture has been found to be a particularly powerful tool in both expressing and recognizing emotion, and the body language portrayed by a posture can serve as a rich source of information that can reveal the goals, intentions and emotions of others". ***C, E, C&E***	Participants had to choose which emotion and action best represented how the character was feeling and performing, respectively	EQ	EQ scores & percentile accuracy and RT of character's emotion and action attribution
Mier et al (2014)	*Discusses embodied simulation in psychopathy: "*Being able to show an emotional reaction to the fate of someone else is highly dependent on our social-cognitive abilities, such as emotion recognition and Theory of Mind (ToM) that help us understanding the emotional as well as mental states of others".***C**	Task that evaluated whether a preceding statement (e.g., "this person is going to run away" or "this person is angry") fitted the facial stimulus ('*joy*', '*angry*', '*fearful*', '*disgusted*', or *neutral*)	N/A	Performance on statement & face matching task (RT and accuracy)
Horan et al (2014)	N/A - discusses imitation as a foundation of social cognition in the context of impaired emotion processing, social perception, and mentalising in Schizophrenia. ***C&E***	N/A	IRI	IRI scores

Braadbart et al (2014)	Discusses empathy in terms of imitation of emotional expressions and acquiring an understanding of "the relationship between the motor plan for an expression and the underlying emotional state the expresser may want to convey". ***M, C&E***	Participants were required to observe and then imitation the expressions presented	IRI	EQ scores & facial imitation accuracy (derived by two scorers)
Pichon et al (2009)	*No explicit definition provided, however, discusses social dysfunction in schizophrenia, and imitation which* "facilitates understanding the actions and even emotions of others through a 'simulation' mechanism". ***C***	Participants named the emotions of the actor (*explicit*)	N/A	Recognition of emotion performance
Alaerts et al (2014)	*No explicit definitions provided.* ***C***	Participants indicated whether the PLD showed a different '*emotional state*' to another PLD clip. A control condition was included (identifying colours changes in the PLDs)	N/A	Correct RTs & accuracy rates of the emotion recognition ask
Flourney et al (2016)	"Empathy is frequently defined as an affective response that is similar to what the other is feeling. This shared experience of emotion, sometimes called affective resonance, can produce sympathy or feelings of empathic concern, widely considered to be important motivators of prosocial behaviour". ***E***	N/A	IRI (only EC & PD subscales)	IRI scores
Woodruff et al., 2011	"Empathy involves various subcomponents, of which there are PT, empathic concern and personal distress to name a few Empathy can occur once the observer becomes aware that the source of first-person thoughts and/or feelings is the other person". ***C&E***	N/A	IRI	IRI scores

Note. ***C*** Cognitive empathy assessed; ***E*** Emotional empathy assessed; ***M*** Motor empathy assessed; ***C&E*** Acquired data consisting of both cognitive and emotional empathy. These were determined based on the method used to measure empathy.

Table 4. Summary of Methodological Approach to Eliciting and Measuring Mirror Neuron Activity Across Studies

Authors (year)	Index of Neural Activity	Action Observation Stimuli	Stimulus Property	Stimulus Type	Control/Baseline Type	Action Observation/ Execution
Schulte-Ruther et al (2014)	fMRI	Facial expressions	Static	Natural	Fixation (*1sec*) & control task (perceptual decision making on the width of stimulus faces)	Passive
Ihme et al (2014)	fMRI	Facial expressions	Static	Natural	Fixation (*800ms*)	Passive
Lenzi et al (2013)	fMRI	Pictures of children aged from 6 - 12 months (expressions of '*joy*', '*distress*', and '*neutral*')	Static	Natural	'Null events', fixation cross, and an inter-trial interval (*50ms*; jittered)	Both
Woodruff & Shelley (2013)	EEG	Black and white videos of; (1) hand laying at rest and (2) same hand shown elevated with the index finger and thumb tapping together	Dynamic	Natural	Participants were shown a blank 5 × 5 grid with a red fixation cross in the centre cell and instructed to visualize each letter of the alphabet, either capital or lowercase, inside the grid, one at a time, once the block began.	Both (*active = mental imagery*)
Libero et al (2014)	fMRI	Static black and white stick figure characters depicting different body postures ('*jumping*', '*throwing*' [physical], and '*upset'*, '*happy*' [emotional])	Static	Stylized	Physical condition & fixation baseline (*24sec*)	Passive

Ferri et al (2014)	fMRI	3 sets of colour videos depicting actors (male and female) (1) grasping objects while facially expressing '*anger*', '*happiness*', or '*no emotion*' (2) just faces showing an actor expressing *anger*', '*happiness*', or '*no emotion*' (2) and only hand actions (*no face*)	Dynamic	Natural	Black fixation (*2000-5000ms*) and '*neutral*' conditions (control)	Passive (*active trials only included to ensure attention*)
Babiloni et al (2012)	EEG	Video (and audio) recording of own music performance in ensemble ('*Observation*') & playing music piece ('*Execution*')	Dynamic	Natural	Resting state (eye-open; *4 mins*) & control condition (observation of video in which they turned pages of a score on a lectern)	Both
Cooper et al (2012)	EEG	3sec video clips of people '*yawning*' or '*haping*' (opening & closing mouths similarly to yawning)	Dynamic	Natural	Fixation (*2000ms*) & blank screen post stimulus (*5000ms*)	Passive
Milston et al (2013)	EEG	Videos of a hand executing actions using the same apparatus used by participants in the execution phase	Dynamic	Natural	Inter-stimulus interval (*500ms*) & inter-trial interval (*500ms*)	Both
Anders et al (2012)	fMRI	5 second clips positive facial gestures	Dynamic	Natural	Scrambled faces & fixation (*8.8s*) & additional baseline after every 6th video block (*33.6 sec*)	Both (*asked to either express or observe the facial gesture*)
Jabbi et al (2015)	fMRI	Viewed facial cues of both still pictures and videos ('*fear' or 'happiness' or 'neutral*')	Dynamic	Natural	Neutral condition	Passive
	MEG	As above	Both	Natural	Neutral condition	Passive

Hoenen et al (2013)	EEG	Video of actor picking up a glass of water, drinking, and putting the glass back on the table	Dynamic	Natural	Baseline (first frame of video depicting actor in resting position; *8sec*) & O1, Oz and O2 recording to control for alpha activity	Both
Lepage et al (2010)	TMS	Videos showing: (1) a rapid movement of the index-finger of the right hand and (2) an immobile hand	Dynamic	Natural	Grey screen (*7sec*) & baseline of 10 MEPs during observation of fixation cross	Passive
Moore et al (2012)	EEG	Photos of faces ('*happy'* and '*disgusted'*)	Static	Natural	One block of static visual white noise images, neutral block (images of buildings) & 'non-empathy' condition	Passive
Bernier et al (2007)	EEG	Observed an adult gripping a manipulandum with thumb and index finger.	Dynamic	Natural	Inter-trial interval (*7sec*) & resting EEG condition	Both
Lepage et al (2014)	TMS	4 videos of a hand grasping a ball (1), a hand performing a grasp movement without object (2), a flat hand moving forward (3), a still flat hand (4)	Dynamic	Natural	10 baseline MEPs acquired during observation of fixation cross & Inter-stimulus interval (*6sec*)	Passive
Perry et al (2010)	EEG	Video clips of point-light displays (PLDs) of continuous biological motion (human walking figure) depicting: '*sad'* or '*happy'* expressions and '*approaching'* or '*retreating'*	Dynamic	Natural	Non-biological movement baseline condition & control sites O1, Oz and O2	Passive

Enticott et al (2008)	TMS	Hand movement observation (*rest, goal directed, continuing*, and *meaningless*)	Both	Natural	Baseline (rest)	Passive
Enticott et al (2011)	TMS	Short videos of two hands (1) motionless (static); (2) L & R hands of same person clasp together; (3) L & R hands from diff people clasp together; (4) L & R hands from same person release a clasp; L hand from person 1 moves to touch person 2 R hand - person 2 moves away (R), avoiding touch from person one (L)	Both	Natural	Black screen (10 baseline MEPs acquired) & black screen between clips (*2sec*)	Passive
Jola et al (2012)	TMS	Observed videos of (1) ballet performance, (2) Indian dance performance (3) and an acting control condition (bodily actions such as clapping, stamping, and hand-fisting)	Dynamic	Natural	Control condition & baseline CSE during rest (eyes-closed)	Passive
Andrews et al (2015)	TMS + EEG	Videos of: (1) 2 static hands or same or different people; (2) hand reaching and grasping a mug (3) hand pantomiming clasping a mug; (4) two interactive movement between two different people and (5) carried out by the same person	Both	Natural	Blank screen (*2sec*) & static hands (control)	Passive
Mehta et al (2014)	EEG	(1) Live observation of experimenter's hand performing grasping actions; (2) virtual observation where participants viewed a video of a hand performing grasping action, and (3) static image of a hand at rest	Both	Natural	Static rest condition	Passive

Brown et al (2016)	EEG	Video clips of two people (1st person or 3rd person point of view, facing each other, transferring coins from one bowl to other bowl labelled "+" (*reward*), "-" (*negative*), and "0" (*neutral*).	Dynamic	Natural	N/A	Passive
Bernier et al (2013)	EEG	Videos of hand gestures and face gestures	Dynamic	Natural	N/A	Both
Silas et al (2010)	EEG	Viewed movements (button press) performed by experimenter	Dynamic (live)	Natural	Inter-stimulus interval (*1.5-2.5sec*) & baseline of non-biological movement resting state (pre and post)	Both
Makhin et al (2015)	EEG	Conditions included: action execution (rhythmically moved mouse around), observation (observed researcher moving mouse around), imitation (imitated researcher's movement), and auditory (perceived sound of researcher's movement)	Dynamic (live)	Natural	Baseline (observing unmoved mouse & relaxing with eyes closed)	Both
Woodruff et al (2016)	EEG	Photographs of '*happy'*, '*sad'*, '*angry'*, and '*neutral'* faces	Static	Natural	Control condition (age estimation block)	Passive
Moore et al (2016)	EEG (eLORETA)	Both video and still images of faces depicting '*angry'*, '*disgusted'*, '*fearful'*, '*joyful'*, *'sad'*, '*surprised'*, '*contempt'*, '*proud'*, '*embarrassed'*, and '*neutral'* expressions	Both	Natural	Fixation period (*10sec*) & control condition (word matching task)	Passive
Schulete-Ruether et al (2011)	fMRI	Morphed photos of '*happy'*, '*sad*', and '*neutral'* male facial expressions (at either low or high intensity)	Dynamic	Natural	Fixation cross (*1sec*) & control condition (perceptual decision of width of neutral faces)	Passive

Castelli et al (2010)	fMRI	RMET task	Static	Natural	Inter-stimulus interval with fixation cross (*5sec*) & control condition (gender identification control task)	Passive
Mazzola et al (2013)	fMRI	Video clips showing the following actions:(1) arm grasping an object on a table; (2) a person with a *neutral* expression grasping an object; (3) '*joy'* or '*angry'* person grasping an object; and (4) '*joyful'* or '*angry'* dynamic face without grasping action	Dynamic	Natural	Inter-stimulus interval (*1810ms*) & 'null events' (*2700ms*)	Passive
Hadjikhani et al (2014)	fMRI	2sec video clips of '*painful'* and '*neutral'* facial expressions	Dynamic	Natural	Fixation cross (*6sec*)	Passive
Likowski et al (2012)	fMRI	Avatar facial emotion expressions of '*happy'*, '*sad'*, '*angry'*, and '*neutral'* emotions	Static	Stylized	Fixation cross (*2000ms*) & inter-trial interval (*250ms*)	Passive
Schulte-Ruther et al (2008)	fMRI	Synthetic emotional faces expression fear or anger	Static	Stylized	Fixation cross (*1.2sec*) & control condition (gender and age decision task of neutral faces)	Passive
Kaplan et al (2006)	fMRI	3sec videos of : background scene with 3 configurations '*context'*, '*drinking context'* & '*cleaning context'*, with two variations: a '*precision grip'* and a 'whole-hand *prehension'*. Finally, these videos consisted of '*congruent'* and '*incongruent'* conditions.	Dynamic	Natural	Fixation period *(12-14sec*)	Passive

Moriguchi et al (2009)	fMRI	4sec videos depicting object related hand actions & artificial hand movement control condition	Dynamic	Natural	Fixation cross (*4sec*) & control condition	Passive
Mier et al (2010)	fMRI	Pictures of faces depicting emotional expressions of '*joy'*, '*anger'*, '*fear'*, '*disgust'*, or '*neutral'*	Static	Natural	Inter-stimulus interval (*2.5-5.5sec*)	Passive
Lee et al (2014)	fMRI	Face stimuli expressing '*happiness'*, '*sadness'*, and a '*meaningless'* expression	Static	Natural	'Null event' conditions (*1.25 - 10sec*) & neutral cartoon face	Both
Hooker et al (2008)	fMRI	Videos of social scenes, with interacting characters	Dynamic	Stylized	Inter-trial interval (*2-6sec*) & rest period (*20sec*)	Passive
Schulte-Ruther et al (2007)	fMRI	Synthetic emotional faces expression '*fear'* or '*anger'*. Participants were required to either concentrate on own emotions emerged from stimuli (*self*) or emotional state expressed by the stimulus face (*other*)	Static	Stylized	Fixation cross (*1.2sec*) & control condition (gender and age decision task of neutral faces)	N/A
McCormick et al (2012)	EEG	Videos of: two bouncing balls; person moving their right hand (opening and closing from the palm, with fingers and thumb held straight); movement of own right hand; and observation of a "live" person (researcher) moving his right hand in same manner has hand movement video conditions	Dynamic	Natural	Baseline, open-eyes task observing visual white noise	Both

Hooker et al (2010)	fMRI	Scenes of characters expressing facial emotions that are based on their belief of what is occurring in a social scene. Characters have different beliefs about what was happening in each scene. At least one character had full knowledge of what is happening ("*true belief*") and one having only partial knowledge or a misunderstanding of what is taking place ("*false belief*"). Subjects classified changes in the scenes as "*social*" (change in characters belief and feelings, "*physical*" (physical change not resulting in change in characters beliefs), and "*no change*"	Static	Stylized	Rest period (*20sec*) & jittered inter-trial interval (*4-6sec*) & 'physical change' condition	Passive
Zaki et al (2009)	fMRI	Videos of another ("*target*") person recalling positive and negative autobiographical events	Dynamic	Natural	Fixation (*2sec*)	Passive
Greimel et al (2010)	fMRI	Photographs of boy/adult male faces expressing '*happiness'*, '*sadness'*, and '*neutral'* emotions	Static	Natural	Fixation cross (*.49 - .95sec*) & control condition (judged width of neutral faces as "*thin*", "*average*", or "*wide*")	Passive
Kana & Travers (2012)	fMRI	Stick figure characters depicting actions (i.e., '*running'*. '*swimming'*, '*pushing'*, '*cart-wheel'*, '*handstand'*, '*sitting'*, '*stretching'*, and '*throwing'*) with emotional content (i.e., '*sad'*, '*happy'*, '*scared'*, '*upset'*, '*relaxed'*, '*confused'*, '*excited'*, '*tired'*, and '*pained'*)	Static	Stylized	Fixation cross (*24sec*)	Passive
Mier et al (2014)	fMRI	Pictures of faces depicting emotional expressions of '*joy*', '*anger'*, '*fear'*, '*disgust*', or '*neutral'*.	Static	Natural	Inter-stimulus interval (*0.5-3.5sec*)	Passive

Horan et al (2014)	fMRI	Videos displaying either an index finder movement or a middle finger movement & faces '*expressing'*, '*happy'*, '*sad'*, '*angry'*, and '*afraid'* emotions, as well as word stimuli. Participants either *observed* the stimuli, *imitate* the movement/facial expression of the stimuli, or *execute* the movement/expression described by each word	Dynamic	Natural	Rest period (8s*)*	Both
Braadbart et al (2014)	fMRI	Face stimuli expressing '*sadness'*, '*anger'*, '*surprise'*, '*fear'*, '*happiness'* and '*disgust'* from the "Sadness-Anger-Surprise & Fear-Happiness-Disgust Triangle".	Dynamic	Natural	N/A	Passive
Pichon et al (2009)	fMRI	3s videos of actors performing the action of opening a door in front of them, reacting to a specified encounter ('*fear'*, '*anger'*, '*neutral'*) and closing the door again. Faces of the actors were blurred, only body expression was available.	Dynamic	Natural	'null events' (*5sec*)	Passive
Alaerts et al (2014)	fMRI	PLDs portraying human actions ('*walking'*, '*jumping'*, '*kicking'*) that expression bodily emotional states ('*anger'*, '*happiness'*, '*sadness'*, or '*neutral'*) in yellow bordered and blue bordered movie clips	Dynamic	Stylized	Fixation block (*16sec*)	Passive
Flourney et al (2016)	fMRI	Facial expressions depicting '*angry'*, '*fearful'*, '*sad'*, '*happy'*, and '*neutral'* expressions	Static	Natural	Inter-trial interval *(0.5 - 1.5sec*)	Passive
Woodruff et al (2011)	EEG	Male hand repeatedly tapping the forefinger and thumb together (observed) & participant replicating hand actions observed (executed)	Dynamic	Natural	Rest condition (observed hand at rest)	Both

3.2. Demographics and Descriptive Characteristics

Sample sizes of studies ranged between 8 and 56 participants ($M_{sample\ size}$ = 25.16, *SD* = 10.34,). These samples consisted of relatively young participants (*M* = 27.67 years, *SD* = 10.14; range = 6.40 – 52.40 years), and, where indicated (6 studies), the average years of formal education was 14.01 (*SD* = 2.11). Overall, there was approximately an equal distribution of sex (51:49; [M:F]), and 52% of studies that reported handedness consisted of right-handed participants. Only four of the included studies recorded ethnicity, and of those, the majority of samples were Caucasian, followed by biracial participants (see table 2 for summary).

3.3. Measurement of the Mirror Neuron System

The majority (53.85%, *n* = 28) of included studies acquired a neural index of the MNS using fMRI, followed by EEG (30.76%, *n* = 16) and TMS (9.61%, *n* = 5). Only 2 studies used MEG and eLORETA, and 1 used combined TMS and EEG. In terms of eliciting a mirror neuron response, 24 studies used facial expressions, 20 relied on hand gestures, and 8 used body expressions and/or point light displays. Of these stimuli, natural (that is, non-animated) were more commonly used (86.53%; *n* = 45), relative to stylized images (13.46%; *n* = 7). Regarding stimulus properties, dynamic images were more frequently used (55.76%; *n* = 29) than static (32.69%; *n* = 17) stimuli, with some presenting both types (11.53%; *n* = 6). Additionally, these stimuli were more often passively observed by participants, with only some studies also including an execution component. All but three studies described the inclusions of control conditions. Most studies (65.31%; *n* = 32) consisted of more than one type of control condition or comparative baseline, including the use of a fixation cross (*n* = 22), a 'rest' condition (*n* = 11), inter-trial intervals (*n* = 14), 'null' events and blank screens (*n* = 8), white noise (*n* = 2), and other control conditions (e.g., judging width of faces as opposed to emotion, neutral images of buildings, or control sites such as O1, O2, O3 [EEG]; *n* = 20; refer to table 4 for summary).

3.4. Definitions and Measurements of Empathy

Descriptions and measurements of empathy were varied across studies. Most studies considered this concept to be mainly subserved by either cognitive (19.23%, *n* = 10) or emotional (13.46%, *n* = 7) processes or both (59.61%, *n* = 31), with only a handful of studies incorporating a motor component to either their definition or measurement (7.69%, *n* = 4). A "simulation" mechanism (or 'shared experience') was frequently mentioned as an important factor for empathy to occur (54%, *n* = 28). Similarly, the "self-other" distinction (which includes the ability to discriminate between one's own emotional and mental states and those of another), which can serve as a protective mechanism for distressing experiences caused by excessive emotional resonance, was also included in a number of definitions and theoretical conceptualisations. In terms of measurements implemented, only 12 studies (23.07%) combined both self-report and experimental measures, while the majority of studies relied on either self-report or experimental measures independently (76.46%, *n* = 40). Of these, 38.46% (*n* = 20) relied on self-report questionnaires, such as the empathy quotient (EQ), the interpersonal reactivity index (IRI), and the balanced emotional empathy scale (BEES), all of which putatively assess either cognitive or emotional empathy, or a combination of both processes, and are consistently used within the field of social cognition. A commonly used clinical questionnaire was the Toronto alexithymia scale (TAS-20), which measures the ability to identify emotions within the self and within others. Further self-report data were collected through simple rating systems following observation of emotional stimuli.

Experimental, or observable, measures were also used (38.46%, *n* = 20), and included the 'reading the mind in the eyes' task (RMET), and other accuracy scores during facial recognition tasks or social situations. There were no physiological measures of emotional empathy, such as measures of heart rate or galvanic skin response. Finally, a combination of methods was used for the measurement of motor processes, including facial electromyography (fEMG) and facial imitation accuracy, which was typically assessed by

independent and trained coders using facial expression coding systems (see table 3 for summary)

3.5. Quantitative Synthesis of the Relationship between the Mirror Neuron System and Empathy (Motor, Emotional, and Cognitive)

Table 5 presents the results of the meta-analysis between MNS activity and each of the empathy domains.

Table 5. Summary of Meta-Analytic Results by Empathy Component and Method of Mirror Neuron Activity Measurement

Empathy component	Method of Mirror Neuron Activity	*N* (articles)	*N* (associations)	Total *N*	*r* (95% CI)	*p*	I^2
Motor	EEG [a, b]	1	2	15	N/A	N/A	N/A
	TMS	0	0	N/A	N/A	N/A	N/A
	fMRI - IPL	0	0	N/A	N/A	N/A	N/A
	fMRI – IFG [a]	3	7	56	.43 (-.94, .99)	.455	91
Emotional	EEG[c]	4	4	54	-.05 (-.55, 47)	.786	68
	TMS[b]	1	3	N/A	N/A	N/A	N/A
	fMRI – IPL[a]	2	3	93	-.31 (-.99, .99)	.561	92
	fMRI – IFG	7	22	205	.30 (-.01, .55)	.053†	72
Cognitive	EEG	7	12	124	-.14 (-.56, .33)	.502	83
	TMS [a]	4	21	37	.08 (-.43, .56)	.645	59
	fMRI – IPL[a]	3	6	82	.34 (-.72, .92)	.349	84
	fMRI – IFG	6	17	167	.44 (-.00, .73)	.051†	78
Cognitive & Emotional	EEG[a]	3	12	53	.36 (-.54, .88)	.228	56
	TMS[a]	2	5	60	.25 (-.99, .99)	.717	93
	fMRI – IPL[c]	2	2	40	.67 (.47, .83)	<.0001*	0
	fMRI – IFG [a]	2	4	40	.74 (-.94, .99)	.140	35

Note. [a]*df*<4 (underpowered and therefore unreliable). [b]Only one study and therefore not meta-analysable. [c]Standard meta-analysis run (no clustering; only one result per study available); no clustering). †p<.10. *p<.05.

3.5.1. Motor Empathy.

No studies examined the relationship between the MNS using TMS and motor empathy. Similarly, no data examining motor empathy and MNS activity in IPL using fMRI were available. There was one study using EEG to examine the MNS and motor empathy. As such, no meta-analyses could be conducted. Data were available for IFG using fMRI to examine the MNS and motor empathy, however, these results revealed no significant associations.

3.5.2. Emotional Empathy.

No studies measuring the MNS using TMS were conducted alongside a measure of emotional empathy. Studies examining emotional empathy and EEG revealed no significant associations. Studies using fMRI for IPL revealed no significant associations. When looking at IFG (using fMRI), a weak to moderate relationship was revealed (r = .30), with only weak evidence against the null hypothesis for this relationship (p = .053).

3.5.3. Cognitive Empathy.

Using TMS, two studies examined cognitive empathy and results showed no significant relationship. Studies assessing this relationship using EEG and fMRI (IPL) revealed no significant associations (see Table 5). Comparably, IFG using fMRI showed a significant, moderate association (r = .44).

3.5.4. Combined Emotional and Cognitive Empathy.

There was no support for an association between the combination of cognitive and emotional empathy and the MNS using TMS or EEG. Measuring the MNS in the IPL region using fMRI revealed a significant strong positive association (r = .67). Studies examining IFG using fMRI also revealed a strong association (r = .74), however, this was not statistically significant at the nominal alpha = .05.

3.6. Moderation and Sensitivity Analyses

Due to the lack of data (i.e., where data were available and analysable by having >1 study), sensitivity and moderation analyses could not be run for all possible comparisons (that is, for all three components of empathy, the different neurophysiological measures of the MNS, and the methodological differences). Where moderation analyses could not be run, we ran sensitivity analyses to compare original meta-analytic effects to the effects that were obtained from the sensitivity analysis. This was done to assess whether the meta-analytic findings were robust to variation in methodological approaches. It should be noted, however, that whilst some results may be observably different from the original effect, large confidence intervals make it difficult to determine whether the overall effect is reliably different when accounting for the different sensitivity analyses, and should be interpreted with caution.

3.6.1. Stimulus Properties (Static vs Dynamic).

Moderation analyses showed that, when assessing the relationship between emotional empathy and the MNS, specifically using fMRI (IFG), there was a significant difference in effect sizes between the use of static images (r = .47, [0.26, 0.64]) and dynamic images (r = -.02, [-.27, .23]), $t(4.21) = 5.71$, $p < .05$, where static images yielded a moderate positive correlation, while dynamic images resulted in a trivial, negative association. When examining cognitive empathy and the MNS using fMRI (IFG), the difference between the use of static images (r = .53, [-.44, .93]) and dynamic images (r = .33, [-.74, .93]) was not significant, $t(3.97) = 0.69$ $p > .05$. Beyond moderation analyses, sensitivity analysis examining the relationship between motor empathy and the MNS using fMRI (IFG) revealed that the use of static images yielded a small association (r = .27). Relative to the original association (r = .43), these results for motor empathy and the MNS using fMRI (IFG) suggest that the original effect was not robust to the methodological variation of stimulus type. Sensitivity analysis for cognitive empathy and the MNS using fMRI (IPL) were also conducted. Results showed that

the use of dynamic images revealed a moderate relationship (r = .26, [-.99, .99]), which was consistent with the original meta-analytic effect (r = .35), suggesting that the original finding was robust to methodological differences.

3.6.2. Administration Type (Self-report vs Experimental).

When examining cognitive empathy and the MNS using EEG, there was no significant difference between the use of self-report measures (r = -.17, [-.94, .89]) or experimental measures (r = .36, [-0.96, 0.99]), t(2.23) = -1.20, p>0.05, although the former yielded a negative correlation, while the latter was positively associated. Similar results were found for cognitive empathy using fMRI (IFG) to measure MNS activity. Moderation analyses revealed that the use of self-report (r = .34, [-.42, .81]) or experimental methodologies (r = .49, [-.30, .87]) did not result in significant differences, t(3.84) = -0.80, p>.05. Sensitivity analyses were conducted for cognitive empathy and the MNS using fMRI (IPL), where only experimental measures were analysable. The results showed that the use of experimental measures yielded a moderate relationship (r = .56, [-.51, .95]). This effect is observably different from the original effect (r = .34), suggesting that this finding was not robust to the methodological difference of administration type. Regarding TMS and cognitive empathy, experimental methods yielded a moderate relationship (r = .41 [-21, .79]), while self-report measured demonstrated a trivial, negative association (r = -.15 [-.62, .40]). Moderation analysis revealed that this was significantly different, t (1.69) = - 8.52, p <.05

3.6.3. Type of Stimulus (Natural vs Stylized).

Moderation analyses showed that, for the relationship between emotional empathy and the MNS using fMRI (IFG), there was a difference between the use of natural images (r = .11) and stylized images (r = .52). This difference was not significant, t(4.17) = 2.42, p= .069, but there was only weak evidence against the null hypothesis for this. Sensitivity analyses were able to be run for both motor and cognitive empathy and the MNS using fMRI

(IFG). Results showed that the use of natural stimuli produced no relationship for motor empathy (r = .13, [-.99, .99]), which is different from the original moderate correlation (r = .43), suggesting that the original effect for motor empathy was not robust to the methodological differences in type of stimulus presented, and that the inclusion of stylized images yielded a stronger correlation. For cognitive empathy results for natural stimuli yielded a moderate relationship (r = .35, [-.13, .69]), which was consistent with the original meta-analytic effect (r = .44), suggesting that the original finding was robust to methodological differences. Notably however, these results comprised effects with large confidence intervals, again emphasising the importance of interpreting with caution. No other sensitivity analyses were able to be run due to insufficient data.

4. Discussion

The overall aim of the current study was to provide a systematic review of the existing literature examining the relationship between the MNS and empathy and to provide a meta-analytic estimate of this relationship. A total of 52 studies investigated the association between the MNS and at least one domain of empathy (motor, emotional, or cognitive). Results provide some evidence to suggest an association between the MNS and empathy. Specifically, results showed a significant positive association between fMRI (IPL) and the cognitive/emotional composite, however, no other associations were found between fMRI (IPL) and any other component of empathy. There was weak evidence for a relationship between fMRI (IFG) and cognitive empathy (p = .051) and emotional empathy (p = .053), however, this was not the case for motor empathy or the cognitive/emotional composite. No significant associations were found between EEG and any component of empathy. Finally, there was no evidence for a relationship between TMS any components of empathy, likely owing to the limited number of studies.

While studies varied in their conceptual and methodological approaches, there were some consistencies, with a number of studies (54%, n = 28) considering the role of *simulation*

(or synonymous terms such as “sharing” of states or “contagion”) as important for empathy. Furthermore, we found great variation in the type and properties of stimuli used to elicit MNS activity, and examined their effects on the main meta-analytic findings.

We will first discuss the qualitative outcomes of this review, including the sample characteristics of included studies, conceptualisation of empathy, and methodological approaches to measuring MNS activity and empathy. Next, we discuss the quantitative results of the current meta-analysis, after which we will examine the statistical effects of methodological variations in measuring the MNS. Finally, we highlight limitations and directions for future research.

4.1. Qualitative Synthesis of the Relationship between the MNS and Empathy

4.1.1. Sample Characteristics of Included Studies.

Sample sizes of included studies were, on average, relatively small ($M_{sample\ size}$ = 25.16, SD = 10.34), which is consistent with the recent findings of Button et al. (2013) that suggest such sample sizes are typical in neuroscience research. The implications of low sample sizes are discussed in subsequent sections. Overall, samples most often consisted of young, educated adults, with an equal distribution of males and females, and approximately half of these participants were right-handed. Although 92% (n = 48) of the included studies did not record ethnicity, our data show that, of those that did provide ethnicity data, samples were mainly Caucasian, followed by biracial participants. It is important to take this opportunity to advocate for more recognition and inclusion of ethnicity data (such as country of birth, ethnic affiliation/identity, etc.) in future research. Like sex and age, ethnicity is a critical ubiquitous variable that should be considered in social neuroscience research, particularly when it comes to understanding human interaction and empathy. The ability to empathise with another is suggested to be stronger with those whom one identifies with, and is less available to out-group members without active effort (Gutsell & Inzlicht, 2012). This understanding has some empirical support, specifically, previous research has shown that

both empathy and the MNS response are affected by interpersonal differences, with participants responding more with those who they identify with, and not with outgroups Gutsell & Inzlicht, 2010; Gutsell & Inzlicht, 2012). Neglecting to consider and account for differences in ethnicity within samples when assessing empathy and the MNS may have important implications.

4.1.2. Measurement of MNS Activity.

The majority of studies using fMRI (53.85%, *n* = 28), followed by EEG 30.76%, *n* = 16), and TMS (9.61%, *n* = 5; with one study implementing both TMS and EEG). Only two studies used alternative measures such as MEG and eLORETA. MNS activity was evoked primarily using presentation of facial expressions and/or hand gestures, while fewer studies used depictions of body expressions or point light displays (PLDs). Regarding types and properties of presented stimuli, natural and dynamic images were more frequently used compared to stylized (i.e., non-human) and static images. Additionally, stimuli were more frequently passively observed by participants, with few studies including an action 'execution' condition. Concerning control conditions and baselines incorporated in study protocols, most studies involved more than one control condition. Primarily, studies administered the following: a fixation cross; neutral images such as buildings; 'rest conditions' as forms of comparative baselines; or 'other' control conditions, such as tasks requiring participants to judge width of faces (as opposed to emotions). Overall, these results highlight the large variation in methodological approach when both eliciting and measuring MNS activity. Although these methodologies seem to be effective in acquiring MNS activity, the large variability in methods may contribute to producing mixed findings within the literature, as well as the results of the current review, which will be discussed in detail below.

4.1.3. Conceptualisations and Measurements of Empathy.

Definitions and understandings of empathy were also varied across studies. Most (*n* = 31) studies included both cognitive and emotional components of empathy, as opposed to adopting one single component, though only seven incorporated motor empathy. Although there were variations in empathy definitions, a common theme emerged. A 'simulation' mechanism, or similar terms such as 'shared internal experiences', were discussed or included in many conceptual frameworks and/or definitions of empathy. This was considered at least once with regard to the three components of empathy discussed herein. The notion of a simulation mechanism, which is frequently considered important for empathy, is theoretically consistent with a number of models regarding the potential roles of the MNS in empathy, whereby the MNS may provide this necessary mechanism at each level (motor, emotional, and cognitive). This is also consistent with the summary model provided in this review.

Regarding empathy measurement, few studies (*n* = 12) incorporated both self-report and experimental (or 'observable') measures, with the majority of studies using only one administration type. However, of these, there seemed to be an equal distribution of self-report (*n* = 20) and experimental (*n* = 20) measures. Self-report measures included the EQ (Baron-Cohen & Wheelwright, 2004), IRI (Davis, 1980), TAS-20 (Parker, Michael Bagby, Taylor, Endler, & Schmitz, 1993), and the balanced emotional empathy scale (BEES; Mehrabian, 1996). These all assess cognitive and emotional empathy, both independently (via the use of subscales [EQ, IRI, & TAS-20]) or collectively (total composite scores). Other measures included self-report rating systems following stimulus presentations. Experimental measures were less diverse, with no studies implementing observable measures of emotional empathy, such as galvanic skin response measures or cardiovascular activity, indicating a high reliance on self-report methods. Cognitive empathy, however, was measured using both self-report and experimental methods, including the RMET, facial recognition tasks, and other social inference tests. Motor empathy, as expected, was only measured experimentally (as there are

no self-report measures of mimicry). However, although facial EMG (fEMG) is considered a more robust measure due to its high temporal and physiological sensitivity, only one study used fEMG, with the remainder opting for facial coding systems in assessing facial mimicry. Again, the large variations in methods and reliance on self-report assessments have implications for results and conclusions, which will be discussed below. Sensitivity and moderation analyses of administration methods were only possible for TMS and cognitive empathy. Results revealed a significant difference between experimental and self-report measures of empathy, where the use of experimental methods resulted in a stronger correlation than self-report measures (see results). This suggests that the use of experimental methods are valuable, and adds weight to the notion that relying on self-report measures alone may be limiting. However, as mentioned earlier, these results should be interpreted with caution.

Overall, despite the large variability in definitions and measurements, there were consistencies in what was considered an important underlying mechanism for empathy to occur, and this particular finding lends itself well to the current theoretical models of the potential MNS-empathy relationship, whereby a simulation mechanism may be key. The meta-analytic evidence for this relationship, however, is examined below.

4.2. ***Quantitative Synthesis of the Relationship between the MNS and Empathy***

4.2.1. Motor Empathy and MNS Activity.

Four studies were found to investigate the relationship between MNS activity and motor empathy. Of these, one used EEG and three used fMRI (examining the IFG only). Only one study utilised fEMG, while the remainder used facial coding systems. It is somewhat surprising that very few studies were found to investigate the relationship between MNS activity and motor empathy, given that motor empathy is theoretically most closely linked with the functional "action-observation" outcomes of the MNS; that is, an automatic motor synchronization of motor behaviour in response to observed movements and actions of

another (Gallese, 2009; Iacoboni, 2009). Across the four studies, no evidence was found for an association between putative MNS activity and motor empathy. It is possible that the methods used to acquire motor mimicry may at least partly explain this result. Only one study used fEMG, which is considered a more sensitive measure that is able to detect rapid muscle contractions that occur below the visual detection threshold (Mauss & Robinson, 2009). By contrast, action coding systems, such as the FACS and the mature imitation task, rely on strong and overt displays of emotions and thus may be suboptimal at identifying rapid and visually undetectable facial muscle activity. Facial coding measures may therefore be insufficiently sensitive to assess subtle and rapid facial activity. Nonetheless, the results of the current meta-analysis provide little support for the role of the MNS in motor empathy at this stage.

4.2.2. Emotional Empathy and MNS Activity.

A total of fourteen studies investigated this relationship, with two studies using EEG, one study using TMS (<2 studies; not analysable), two studies using fMRI (IPL), and seven studies using fMRI (IFG). All included studies used self-report measures of empathy, and no experimental methods. There was weak evidence (i.e., $p = .053$) for a positive, moderate association between IFG activity and emotional empathy. However, there was no evidence for an association between emotional empathy and MNS activity using EEG or fMRI (IPL). Notably, the meta-analysis examining the relationship between emotional empathy and IFG activity comprised a larger pooled sample size ($N = 205$), relative to EEG and fMRI (IPL; $N = 54$ and $N = 93$, respectively; see Table 5), which may explain the higher power to detect true effects in the former analysis. In any case, this result provides preliminary evidence for a relationship between emotional empathy and MNS activity in the IFG region of the MNS. This is consistent with previous models, whereby the IFG region of the MNS is thought to be involved in emotional contagion, and considered an important structure of emotional empathy (Shamay-Tsoory, Aharon-Peretz, & Perry, 2009). Regarding the studies using EEG and fMRI

(IPL), for which there was no evidence to support a relationship, it is plausible that the 'core' VM-MNS may not play a direct role in emotional empathy, and that the 'mirroring or 'neural resonance' previously described in the literature may be more heavily linked to the "extended MNS" and its connections to regions of the brain involved in emotion processing (Carr et al., 2003).

Additionally, methodology may contribute to the varied results concerning this relationship. None of the included studies acquired observable measures of emotional resonance, such as cardiovascular responses, electrodermal activity, or other autonomic-related changes in response to emotional stimuli. While self-report measures provide valuable insight into the experience of participants, it has been suggested that these data may not be sufficient in determining the extent of emotional resonance experienced outside of conscious awareness and understanding (Ciuk, Troy, & Jones, 2015). It is also thought that self-report questionnaires are subject to social desirability biases, and may reflect 'cognitive appraisal' of emotion, as opposed to the emotional experience itself (Ciuk et al., 2015; Larsen, Berntson, Poehlmann, Ito, & Cacioppo, 2008). Methods assessing changes in physiology in response to emotional stimuli address this limitation (Mauss & Robinson, 2009). While physiological measures are not without their limitations (such as costs associated with data collection, and specificity of changes in autonomic activity), they provide relatively objective measures of emotional arousal that can be used in conjunction with self-report data to provide a rich measure of emotional experience. Finally, it is possible that the studies showing no relationship may have lacked statistical power and precision to detect smaller effects. The results of the current meta-analysis are somewhat inconclusive, and it remains unclear how the MNS relates to the experience of emotional empathy.

4.2.3. Cognitive Empathy and MNS Activity.

A total of 20 studies were found to examine cognitive empathy and MNS activity. Of these, seven used EEG, four used TMS, and nine used fMRI ($N = 3$ for IPL, $N = 6$ for IFG).

When examining specific results, there was evidence for a positive, moderately sized relationship between cognitive empathy and MNS activity using fMRI, although the evidence for this was weak (p = .051; see Table 5) . There was no evidence to support a relationship between cognitive empathy and IPL (fMRI), TMS, or EEG activity. Where possible, sensitivity and moderation analyses for this component suggested that these effects were robust to methodological differences, where no significant differences were found between the administration type (self-report or experimental). However, this was not the case for fMRI (IPL), where sensitivity analysis indicated that the use of experimental measures yielded a stronger correlation than the original effect, again, emphasising the importance of obtaining diverse measures.

The current meta-analysis provides weak evidence for a relationship between cognitive empathy and IFG (fMRI) MNS activity. This is consistent with previous models, where the IFG was proposed to be important for emotion recognition (Shamay-Tsoory et al., 2009). It should be noted, however, that these previous models categorised 'emotion recognition' as *emotional* empathy, however, we consider this process *cognitive* empathy, seeing as it requires mentalising aspects (i.e., consciously identifying another's emotions/emotional states) that are dissociable from emotional resonance.

The current results suggest the MNS may play a role in cognitive empathy, which provides some empirical support for previous theories, such as simulation theory, predictive coding and associative learning accounts of the MNS. In other words, the ability to infer and understand the mental states of others may be facilitated by 'embodied simulation', whereby congruent internal representations within the observer are activated during interactions in order to facilitate increased understanding of the observed agent (Gallese, 2001, 2008, 2014). These internal representations, which accumulate over time via associative learning (through prior experience and social exchanges), are provided by the action-observation properties of the MNS, where mental states of others are inferred through direct observation of action

(Brown & Brüne, 2012; Friston, 2010; Frith & Frith, 1999; Heyes, 2012; Heyes & Ray, 2000). Essentially, these theories collectively postulate that cognitive empathy can occur based on internal models acquired via the interaction between sensory inputs (i.e., observed behaviour and observed motor/physical information) and prior knowledge or experience (i.e., stored internal representations through social experiences; Brown & Brüne, 2012). However, similar to emotional empathy, the current results also show no relationship between cognitive empathy and the IPL, suggesting that perhaps the IFG may play a more active role in cognitive empathy. Overall, the results of this meta-analysis are relatively consistent with previous theories, suggesting that the MNS may potentially be one neurophysiological mechanism subserving cognitive empathy.

4.2.4. Cognitive and Emotional Empathy Composite and MNS Activity.

As mentioned earlier (see method subsection 'data extraction'), we also combined cognitive and emotional empathy for studies that did not examine components of empathy separately (i.e., where total composite scores were reported). A total of nine studies were found to examine MNS activity and a cognitive and emotional composite. These studies used EEG ($N = 3$), TMS ($N = 2$), and fMRI (IPL, $N = 2$; IFG, $N = 2$). Results for the cognitive/emotional composite and MNS activity displayed weak to strong positive associations across MNS methods, but only the association with fMRI (IPL) was found to have strong support ($p < 0.0001$). However, it should be noted that only two associations were available for analysis and should be interpreted with some degree of caution, suggesting that further studies are necessary to fully elucidate this relationship.

Overall, there is some evidence to support an association between MNS activity and empathy, however, the extent of this relationship remains unclear and inconclusive, perhaps owing to methodological variations, among other limitations.

4.3. Methodological Variations in the Measurement of MNS Activity

The current review assessed differences in methodological approaches to measuring the MNS, specifically with regards to type of stimulus (natural or stylized) and stimulus properties (dynamic or static). We found that studies typically used either dynamic images (55.76%; *n* = 29), or static (32.69%; *n* = 17), or a combination of both (11.53%; *n* = 6) to activate the MNS, however, the reasons for the choice of stimuli were not explicitly justified. Where possible herein, moderation and sensitivity analyses were run to test whether these variations had an effect on the relationship between the MNS and empathy overall, as well as by subcomponent. Our results showed that the type of stimulus displayed had an effect on MNS activity and emotional and motor empathy, but not on cognitive empathy. Results were varied, with some suggesting that static images yielded stronger correlations (e.g., emotional empathy and MNS activity using fMRI [IFG]), while others suggested the opposite (e.g., motor empathy and MNS activity using fMRI [IFG]). However, it is crucial to note that all moderation and sensitivity analyses presented here should be interpreted with extreme caution, considering the small number of comparisons that could be made. It should be noted that static images may be effective in eliciting putative MNS activity if images used contain *implied* movement (dynamic information within static images, or 'implicit motion'; Proverbio et al., 2009; Urgesi et al., 2006). While speculative, this may partially account for these varied results. However, the current study did not differentiate between 'sub-types' (i.e., implied vs not) of static images, and therefore is inconclusive in this regard. Future studies should consider recording type of static images, as this may influence the extent of MNS activity evoked.

Regarding the types of stimuli used to elicit MNS activity, most studies utilised images containing natural human faces (86.53%; *n* = 45), with only seven (13.46%) studies using stylized or animated images such as cartoons and/or avatars. It has been previously theorized that the use of unnatural images potentially elicits different neurological responses and behavioural outcomes relative to natural images and that results from human and avatar

data should be interpreted with caution, as they may not be directly comparable (de Borst & de Gelder, 2015; Mori, 1970; Sarkheil et al., 2012). Using moderation analyses, we tested whether using stylized or natural images had an effect on the relationship between MNS activity and empathy. We found, for emotional empathy, weak evidence (p = .06) to suggest a difference between the use of stylized and natural images, whereby stylized images produced a stronger relationship. Only sensitivity analyses were able to be run for the other components of empathy. For motor empathy, the original effect was not robust to the methodological differences in type of stimulus presented, and results showed that the inclusion of stylized images yielded a stronger correlation, relative to natural stimuli. However, this was not the case for cognitive empathy, where the original meta-analytic effect was robust to methodological differences. Again, these results should be interpreted with caution due to the limited data available for analysis. It remains unclear whether the use of stylized or natural images have an effect on the relationship between the MNS and overall empathy, however, the present data set indicated limited evidence of an effect.

4.4. Limitations & Recommendations for Future Research

The literature varies regarding definitions and subsequent measurements of empathy and is perhaps over-reliant on subjective/self-report measures. These limitations have implications on current attempts to understand and investigate the relationship between this construct and the MNS, as these variations and inconsistencies may contribute to the mixed and inconclusive findings within the literature. The issue concerning varied understandings and conceptualisations of empathy is a difficult problem to address, as there is yet to be a comprehensive and empirical examination of this complex construct, where an integrative and refined understanding of empathy can be generated, validated, and potentially agreed upon within the research area. The current review and meta-analysis highlights the need for such investigation, for a number of reasons. First, the field of empathy research is growing rapidly, so developing an agreed upon and valid model for empathy is necessary to ensure

consistency in future research. Second, developing an understanding of empathy that is crucial for healthy social functioning will not only stimulate future research, but can also have important clinical implications, as numerous psychological and psychiatric conditions are characterised by impairments in the ability to empathise (such as autism spectrum disorder, schizophrenia, and borderline personality disorder). Third, while there are current attempts to "train" or *augment* empathy (for example "clinical empathy training" to enhance patient outcomes, and in school programs to prevent bulling behaviour), there are no evidence-based methods of doing so. Finally, and perhaps most relevant to the current meta-analysis and imperative to the previous points, a definitive empathy model would potentially provide new insights and allow for a more thorough, robust, and comprehensive investigation into the relationship between empathy and the MNS, in turn facilitating more conclusive findings and providing a new platform for future studies. This may greatly improve our understanding of the relationship between empathy and the MNS, and can potentially lead to more effective methods of training and enhancing empathy in both community and clinical settings.

Additional limitations include the variations in MNS elicitation strategies and measurements. In terms of its measurement, only one of the 52 included studies implemented both EEG and TMS concurrently. As discussed earlier, a previous study reported no correlation between these two methods of measuring mirror neuron activity, and it has been suggested that relying solely on one may not be sufficient (Lepage et al., 2008). It is not clear why there was no direct linear relationship between two methods thought to measure the same system. One explanation suggested that these methods tap into different neurophysiological aspects of the MNS, where TMS produces motor-related activity, whilst EEG reflects sensorimotor properties of MNS activity. It has therefore been recommended that both be used concurrently (Lepage et al., 2008). In addition, there was great methodological variation in techniques used to elicit MNS activity throughout studies, with

little consistency. The best methods for measuring and/or eliciting MNS activity remain uncertain, and there is currently no universally agreed upon or empirically validated 'gold standard'. Future research should aim to develop more clear understandings regarding the reliability and validity of the methods used to elicit a MNS response, as well as techniques used to acquire a mirror neuron response. Additional techniques, such as the use of concurrent eye-tracking, could also assist with accounting for other possible limitations not currently considered, such as attentional or perceptual factors. The development of more robust measures can potentially allow for methodological consistencies across laboratories examining the MNS, and hopefully lead to more reliable outcomes. Based on the current meta-analysis, it remains unclear which stimulus type or property is most effective in eliciting MNS activity, highlighting the need for further investigation.

Further to this, it is important to note that the main results of the current meta-analysis are likely to be affected by underpowered studies and should be interpreted with caution. In fact, as revealed in a previous meta-analysis (Button et al., 2013), small sample sizes and low statistical power are typical in neuroscience research (between approximately 8% and 31%) and have been previously described as "endemic" (Button et al., 2013). Our meta-analyses demonstrate findings consistent with those of Button et al. (2013), where the average sample size was 25 and ranged between eight and 56 participants). Furthermore, I^2 values for a number of comparisons were high, suggesting considerable heterogeneity due to between study variation and inconsistencies. However, given the large confidence intervals, it is difficult to interpret heterogeneity for this meta-analysis. In light of these limitations, future research would benefit from larger sample sizes in order to ensure adequate power. This can be achieved by performing *a priori* power calculations using prior literature to determine appropriate sample sizes (Button et al., 2013).

A further key limitation is the file-drawer problem, or publication bias, which has been described as a critical issue that may compromise the reliability of meta-analytic results

(Dalton, Aguinis, Dalton, Bosco, & Pierce, 2012; Rosenberg, 2005; Rosenthal, 1979). The file-draw problem occurs due to selective publication where nonsignificant results are less likely to be published, relative to significant "positive" results, introducing bias into the scientific literature and, subsequently, meta-analytically derived effect sizes (Dalton et al., 2012). Additionally, *published* studies existed where comparisons were found to be statistically non-significant were described in-text, but the relevant test statistics were not reported. In other words, the statistical estimates of non-significant effects are not reported in a number of studies included in this review, resulting in missing values and adding to the issues surrounding biased meta-analytic results. Due to this, studies included in meta-analyses, including the current, may not reflect all existing studies conducted examining the relationship of interest and meeting inclusion criteria, and as a consequence, meta-analytic findings may not be representative (Dalton et al., 2012; Viswesvaran, Barrick, & Ones, 1993). It should be noted that the inclusion of 30 correlations received from only seven of 41 contacted authors resulted in significant changes in some of our results, particularly the cognitive empathy and TMS association, which changed from a moderate association approaching significance (r = .41 [-.22, .78], p = .07) to a trivial non-significant relationship (see Table 5). This highlights the instability of incomplete of meta-analyses due to unreported data and emphasises the need to interpret the current results with a degree of caution.

A final limitation worthy of discussion pertains to the inclusion of fMRI studies in this meta-analysis. Methodological approaches (a-priori ROI or peak-voxel) were highly variable across studies, which may have resulted in artificial inflation of the current results due to selection bias within these studies. Although, in the current study, we attempted to categorise each study accordingly in order to meta-analytically account for such variation, this could not be achieved due to the vast number of different approaches, where studies used multiple approaches and therefore could not be differentiated.

In light of the limitations and methodological issues discussed, there are a number of implications and key recommendations for future research. To begin with, future research should aim to provide all relevant statistics (even those that are not statistically significant), including unadjusted effect sizes, and analysis information. These results can be provided in supplementary materials, so that researchers may access them and circumvent bias. As highlighted by Button and colleagues (2013), it should be a key priority in neuroscience research to perform *a priori* power calculation and design studies accordingly to ensure sufficient sample sizes and predictive power, and to also to disclose findings transparently (Button et al, 2013).

5. Conclusions

These caveats notwithstanding, the findings of the current systematic review and meta-analysis suggest that the MNS may play a role in its functioning of empathy. It is clear that there is a need for experimental studies examining the nature of empathy in a comprehensive and integrative manner using a number of subjective and objective research tools, with the aim of developing and refining an empirically validated model for empathy. This model would potentially provide new insights and provide a new platform for future studies.

Despite the controversy and varying accounts of the role of the MNS in empathy, the current meta-analysis provides preliminary evidence of a positive association in some domains of empathy, depending on techniques of MNS elicitation and measurement strategy. One critical omission, however, is the issue surrounding causality, and research is lacking in efforts to directly determine a causal link. Although we have demonstrated an association, it remains unclear whether the MNS is causally involved, or *necessary* for empathy. The current meta-analysis gives rise to a number of follow-up questions that require further investigation, including whether manipulating MNS activity would influence empathy domains directly, or whether it would be mediated or moderated by a different primary

relationship governed by other processes or neural regions. It is possible that the association may be driven by other causes, and a definitive indication of the extent of involvement of the MNS in empathy is yet to be determined.

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